Graphical Abstract

Decarbonizing breweries and distilleries through integrated process and energy system optimization

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Highlights

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- Analyzing feasibility of different decarbonization options in brewery and distilleries industry
- Exploring the potential of synergies among the brewery and distilleries industry
- Performing sensitivity on prices for exploring different options of technology and waste valorization in the industries.
- Analyzing the impact of decision in foreground on the background of the industry.

Decarbonizing breweries and distilleries through integrated process and energy system optimization

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Abstract

This study presents an integrated framework for decarbonizing brewery and distillery energy systems by combining DWSIM process simulation, pinch analysis, mixed-integer linear programming, techno-economic analysis, and life cycle assessment. Three production scenarios are evaluated: beer, whiskey, and combined beer–whiskey production. Heat integration reduces external heating and cooling requirements by 50–75%, although a common pinch temperature of approximately 98.6 °C leaves a residual demand for external heat. Technology selection is evaluated for electricity prices of 0.05–0.50 €/kWh, natural-gas prices of 0.05–0.30 €/kWh, and carbon taxes of 0, 80, and 200 €/tCO_2. Multi-level heat pumps emerge as the most robust option, reducing external final-energy input by 66–79% and energy-related fossil CO_2 emissions from 0.18–0.34 to approximately 0.03 t_{CO_2}/t_{crop}. Solar thermal integration reduces natural-gas consumption by up to 16% but is constrained by seasonal availability and temperature matching. Methanation produces 1.5–1.7 MWh_{CH_4}/t_{crop} of synthetic methane while consuming 2.7–3.1 MWh_{elec}/t_{crop}, making it competitive only at low electricity and high natural-gas prices. Life cycle assessment confirms that heat-pump-based configurations generally reduce carbon footprint and non-renewable energy use, whereas methanation increases non-renewable energy use and water-related impacts to 1.4–1.8 and 1.8–2.1 times their respective baseline values. Similar trends across all production scenarios indicate that technology selection influences environmental performance more strongly than production allocation. The results establish heat pumps as the principal decarbonization technology,

complemented by heat recovery, waste valorization, and context-dependent solar thermal and methanation options.

Keywords: Process Integration, Optimization, Techno-economic modeling, DWSIM, Brewery, Distilleries,

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1. Abbreviations

LCA : Life Cycle Assessment
NRTL : Non-Random Two-Liquid
CHP : Combined Heat and Power
CIP : Cleaning In Place
MILP : Mixed Integer Linear Programming
MER : Minimum Energy Requirement
AMPL : A Mathematical Programming Language
LCIA : Life Cycle Impact Assessment
CC : Climate Change
EQ : Ecosystem Quality Damage
NR : fossil and nulcear energy use
HH : Human Health damage
WF : Water scarcity Footprint
GCC : Grand Composite curve
UGCC : Utility Grand Composite Curve
HP : Heat pump
ST: Solar thermal
BGE: Biogas Engine
WWT : WasteWater Treatment
NG : Natural Gas
TOTEX : Total Expenditure
COP : Coefficient of Performance
BMP: Biomass Potential

2. Introduction

Global final energy consumption reached approximately 360 EJ in 2022, with low-temperature industrial heat accounting for approximately 7.1% of total energy use [1]. At the same time, more than 110 countries have adopted

or announced net-zero targets [2], increasing the need for technically feasible and economically viable industrial decarbonization strategies. Research and policy have traditionally focused on energy-intensive sectors such as iron and steel [3, 4], petroleum refining [5], and chemicals [6]. However, low-temperature industries, including food and beverage processing, also consume substantial amounts of thermal energy and electricity[7]. These industries present distinct decarbonization challenges because production is distributed across many facilities, heating and cooling demands often occur simultaneously, and environmental impacts extend into agricultural and energy supply chains.

Breweries and distilleries provide relevant examples of low-temperature process industries. Both production routes include common operations such as mashing, boiling, fermentation, cooling, and cleaning, while distilleries additionally require energy-intensive separation through distillation [8, 9]. Their overlapping process structures create opportunities for internal and cross-process heat recovery. In addition, fermentation and wastewater treatment generate biogenic CO_2 and organic by-products that can potentially be valorized through anaerobic digestion, combined heat and power generation, or synthetic methane production [10, 11, 12]. These characteristics make brewery and distillery systems suitable case studies for investigating interactions among heat recovery, electrification, renewable heat, waste valorization, carbon utilization, and storage.

Previous studies have investigated process modelling of breweries [8] and distilleries [9], as well as individual decarbonization strategies including heat recovery [13], heat-pump electrification [14], renewable-energy integration [15], carbon utilization [12], wastewater valorization [10], dregs valorization [11], and hydrogen integration [16]. These studies demonstrate the potential of individual technologies, but their interactions are less frequently evaluated within a common process and energy-system framework. In particular, limited attention has been given to the simultaneous consideration of process heat integration, heating and cooling utility selection, temporal resource availability, energy-price variability, carbon taxation, waste valorization, and storage. Furthermore, configurations that reduce operating costs and facility-level fossil CO_2 emissions may shift environmental burdens toward upstream electricity generation, water use, or resource consumption. Consequently, economic optimization and facility-level carbon accounting alone are insufficient for identifying environmentally robust industrial decarbonization pathways.

This study develops an integrated process-to-system framework combining process simulation, pinch-based heat integration, multiperiod mixed-integer linear programming, techno-economic sensitivity analysis, and cradle-to-gate life cycle assessment. Brewery and distillery processes are first modelled in DWSIM to quantify their material, heating, and cooling requirements. The resulting process models are subsequently integrated into an OSMOSE superstructure containing natural-gas furnaces, refrigeration systems, multi-level heat pumps, solar thermal systems, biogas engines, wastewater valorization, biogenic CO_2 methanation, and storage technologies. Three production scenarios are considered: beer production, whiskey production, and combined beer–whiskey production. The combined scenario is used to determine whether differences in process temperature profiles create opportunities for cross-process heat integration and shared utility infrastructure.

The specific objectives of this work are to:

1. quantify the reduction in external heating and cooling requirements achievable through internal and cross-process heat integration;
2. determine the electricity-price, natural-gas-price, and carbon-tax conditions under which heat pumps, solar thermal systems, waste valorization, methanation, and storage technologies are selected;
3. identify the economic conditions governing dregs-storage activation under temporal electricity-price variability; and
4. evaluate whether cost-optimal and low-carbon configurations provide consistent improvements across life cycle impact categories or shift environmental burdens upstream.

The main contribution of this study is the simultaneous connection of process-level thermodynamic integration, multiperiod energy-system optimization, waste and carbon valorization, storage flexibility, and configuration-specific life cycle assessment. Although the framework is demonstrated for brewery and distillery systems, it is applicable to other low-temperature industries characterized by simultaneous heating and cooling demands, variable renewable-energy availability, and recoverable biogenic resources.

3. Methodology

This section presents the overall methodological framework used to assess brewery and distillery operations. The processes are first modelled in DWSIM and subsequently integrated into OSMOSE for pinch analysis and

utility-system optimization. The decarbonization technologies considered, together with the techno-economic and environmental indicators used to compare alternative configurations, are then introduced. Finally, the LCA methodology used to evaluate the environmental performance of the selected configurations is outlined.

Given the similarities between the production processes for beer and whiskey, there is potential for cross-process heat integration. To explore this, the study considers three distinct production scenarios:

- **Case A**: All crop is processed into beer.
- **Case B**: All crop is processed into whiskey.
- **Case C**: The crop is split evenly, with half processed into beer and half into whiskey.

These scenarios enable assessment of the impact of production allocation on energy integration, system optimization, and environmental performance.

3.1. Process modeling and simulation

The process flow diagrams for beer and whiskey production are illustrated in Figure 1. The main unit processes considered in this study include mashing, boiling, fermentation, and subsequent thermal treatments such as cooling and maturation, followed by either pasteurization or distillation. Additionally, auxiliary energy requirements associated with bottle cleaning are included within the system boundaries.

The data for raw material requirements across different unit operations are sourced from [13] the flow rate table is present in TableD.12 . The processes are modeled using DWSIM, which enables the evaluation of mass, energy, and exergy balances for all unit operations. The NRTL property package is considered. The modeling approach for each unit process is described in the following subsections.

For consistency in comparison, both production pathways are modeled using the same input quantity of crops (10.26 ton/h). The beer production process yields approximately 98 m^3/h of beer, whereas the whiskey production process results in a much lower output of approximately 5.8 m^3/h of whiskey, reflecting the concentration effect and separation intensity associated with distillation.

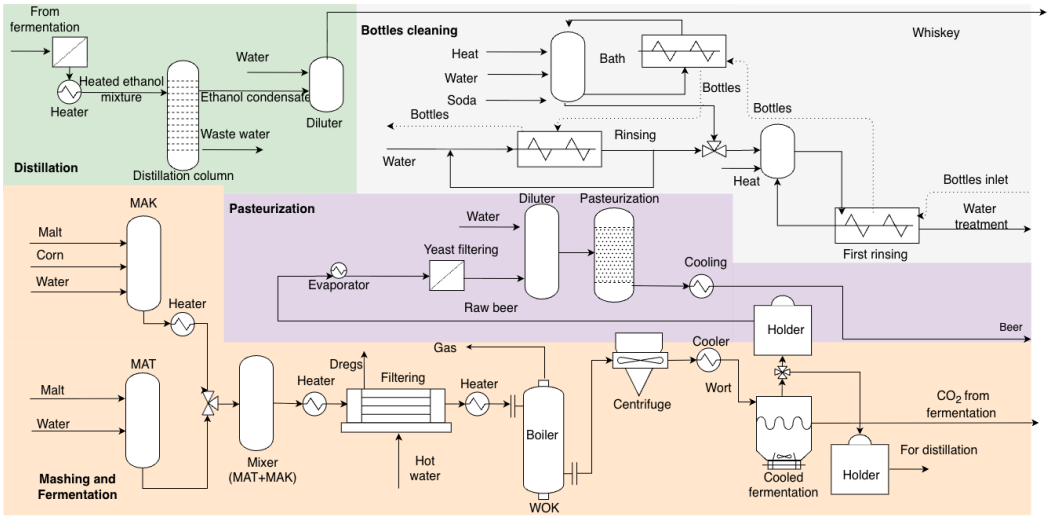


Figure 1: Process flow of beer production and whiskey production

3.1.1. Malting and Fermentation

A variety of crops can be used as feedstock for beer and whiskey production. All organic solid components are represented as maltose to simplify the system. The process begins with the mixing of two streams containing different compositions of maltose and water. The resulting mixture is heated and subsequently combined with hot water. It is then passed through a filtration unit, where solid residues (dregs) are removed as a by-product.

The filtered stream is directed to the WOK unit shown in Figure 1, which is modeled as a conversion reactor. In this unit, maltose reacts with water to form glucose according to the following reaction:

$$\mathrm{C_{12}H_{22}O_{11} + H_2O \longrightarrow 2\,C_6H_{12}O_6} \tag{1}$$

Following this step, the stream is sent to a centrifuge to remove any remaining suspended solids. The clarified liquid is then directed to the fermentation stage, modeled using a conversion reactor. Glucose is converted into ethanol and carbon dioxide according to the reaction:

$$\mathrm{C_6H_{12}O_6 \longrightarrow 2\,C_2H_5OH + 2\,CO_2.} \tag{2}$$

3.1.2. Pasteurization process

Following fermentation, the product stream is first sent to a separation unit equipped with a CO_2 recovery system, where carbon dioxide is removed

from the mixture. The remaining liquid stream, primarily composed of ethanol and water, is then directed to an adjustment step in which water is added to achieve the desired ethanol concentration for beer (ethanol 3-5% by mass).

Subsequently, the adjusted mixture undergoes thermal treatment through pasteurization, where it is heated to ensure microbial stability. The pasteurized beer is then cooled to storage conditions prior to packaging.

3.1.3. Distillation process

After fermentation, the product stream is similarly routed to a separation unit with CO_2 recovery, where carbon dioxide is removed from the mixture. The resulting ethanol–water solution is then fed to a distillation column for ethanol enrichment .

The distillation system in DWSIM is modeled using a shortcut column approach. The light key compound stream, consisting of an ethanol-rich vapor mixture, is subsequently condensed to obtain the final whiskey product(ethanol 33-43% by mass).

3.1.4. Bottle cleaning

The bottle cleaning system mainly aims at preparing the packaging materials (mainly bottles and kegs) before filling it up with beer or whiskey. This process is considered as a continuous operation. The bottles are successively pre-rinsed, washed with soda and rinsed in successive baths again before being filled with beer. The temperature of the baths is maintained by steam injection. The final rinsing is done with the clear water at 15°C. Water of the final bath rinsing is recuperated and sent to the pre rinsing bath, recovering heat form the bottles that are cooled down in the last bath.

3.2. Utilities Integration

All the three cases are energy-intensive processes involving multiple stages, each characterized by distinct heating, cooling, and mechanical or electrical energy demands. Currently these demand are being met with the help of natural gas, water and electricity. This dependence on conventional energy sources results in high operational costs, significant greenhouse gas emissions, and increased vulnerability to fuel and electricity price volatility. In addition, the intermittent nature of renewable energy sources poses challenges for ensuring a consistent energy supply throughout the conversion process.

A more sustainable conversion system can be developed by considering options of low-carbon technologies and resource recovery strategies, as illustrated in Figure 2. The key technologies for such integration are outlined below:

- **Low-Carbon Heating Options** – Natural gas boilers can be replaced or supplemented with heat pumps to provide process heat more efficiently. By optimizing evaporator and condenser temperature levels, it is possible to simultaneously reduce both heating and cooling demands In current analysis an exegetic efficiency of 0.5 is taken for the consideration. In addition, parabolic solar thermal collectors can supply process heat during periods of solar availability, thereby reducing fossil fuel consumption. The irradiance distribution is shown in Figure 3.

- **By-Product Valorization** – Brewer's dregs (spent grains) can be converted into biogas through anaerobic digestion or thermochemical processes. The resulting biogas can be utilized in a CHP unit to generate both electricity and heat on-site. In this case CHP unit is assumed to have an electrical efficiency of 0.41 and thermal efficiency of 0.42.

- **Biogenic Carbon Utilization** – Carbon dioxide (CO_2) emissions from fermentation can be captured and reacted with renewable hydrogen produced via electrolysis (e.g., using alkaline electrolyzer) through the Sabatier process to generate methane (CH_4). This synthetic natural gas can be used as an energy carrier within the system. As CO_2 methanation is an exothermic reaction, the released heat can also be recovered and utilized within the process. In current case 55 $\mathrm{kWh/kg}_{H_2}$ of electrical power is required by the electrolyzer and it produces around 2.5 $\mathrm{kWh/kg}_{H_2}$ the values for the electrolyzer are taken from [17]. The total waste heat released by methanation is around 3.8 $\mathrm{kWh/kg}_{CH_4}$ [18].

- **Energy and Gas Storage** – Storage systems for CO_2, CH_4, and thermal energy can be implemented to enhance system flexibility. CO_2 storage enables controlled methanation, while methane storage supports CHP operation and heating demand. Thermal energy storage helps balance supply and demand, mitigates intermittency of renewable resources, and buffers against market price fluctuations.

- **Efficient Cooling Systems** – Cooling towers can be employed for general heat rejection, while high-efficiency refrigeration units ensure precise temperature control during fermentation and storage stages.

- **Wastewater Treatment** – Wastewater generated from mashing and CIP processes can be treated to produce sludge which in turn produces biogas. Wastewater is assumed to have Total Solid of 0.04, Volatile Solid of 0.742 and the BMP of 407 $\mathrm{Nm}^3_{CH_4}/\mathrm{t}_{VS}$ these values are coming from [19].

- **Dregs Storage** – Storing brewer's dregs provides operational flexibility, allowing their utilization during periods of high utility prices to offset energy costs. The storage can act as a long term storage because of the time required for the anaerobic digestion is 20 days. Cost of dreg storage is taken to be (15 €/ton_{dregs}). This is primarily due to its direct compatibility with existing infrastructure and the absence of additional conversion requirements.

3.3. Optimizing utility integration and problem definition

The utility integration optimization framework in this study is adapted from the methodology proposed by [20], which integrates sequential steps of process modeling, simulation and optimization. The selection among the proposed utility units and process alternatives is formulated as a MILP problem. Given the possibility of both importing energy vectors from the respective grid and generating it internally (e.g. through the valorization of dregs or via the solar thermal plant), the system must balance external energy purchases against internal generation. Therefore, the optimization aims to minimize the overall utility cost by identifying the most economical pathway for minimum grid import. It must also be noted that outcomes entirely depends on the initial number of energy technologies and resources adopted, and the window of the appropriate operating conditions assumed (e.g. heat pump levels, Solar thermal operations).

In the first stage, a comprehensive list of all feasible utility systems, such as the furnace, refrigeration unit, heat pump, and cogeneration system. The computational framework subsequently interfaces with DWSIM to exchange process data and formulate the MILP problem described in Eqs. (1)–(6). This optimization problem seeks to minimize the overall consumption of resources such as water, natural gas, and electricity, depending upon the choice

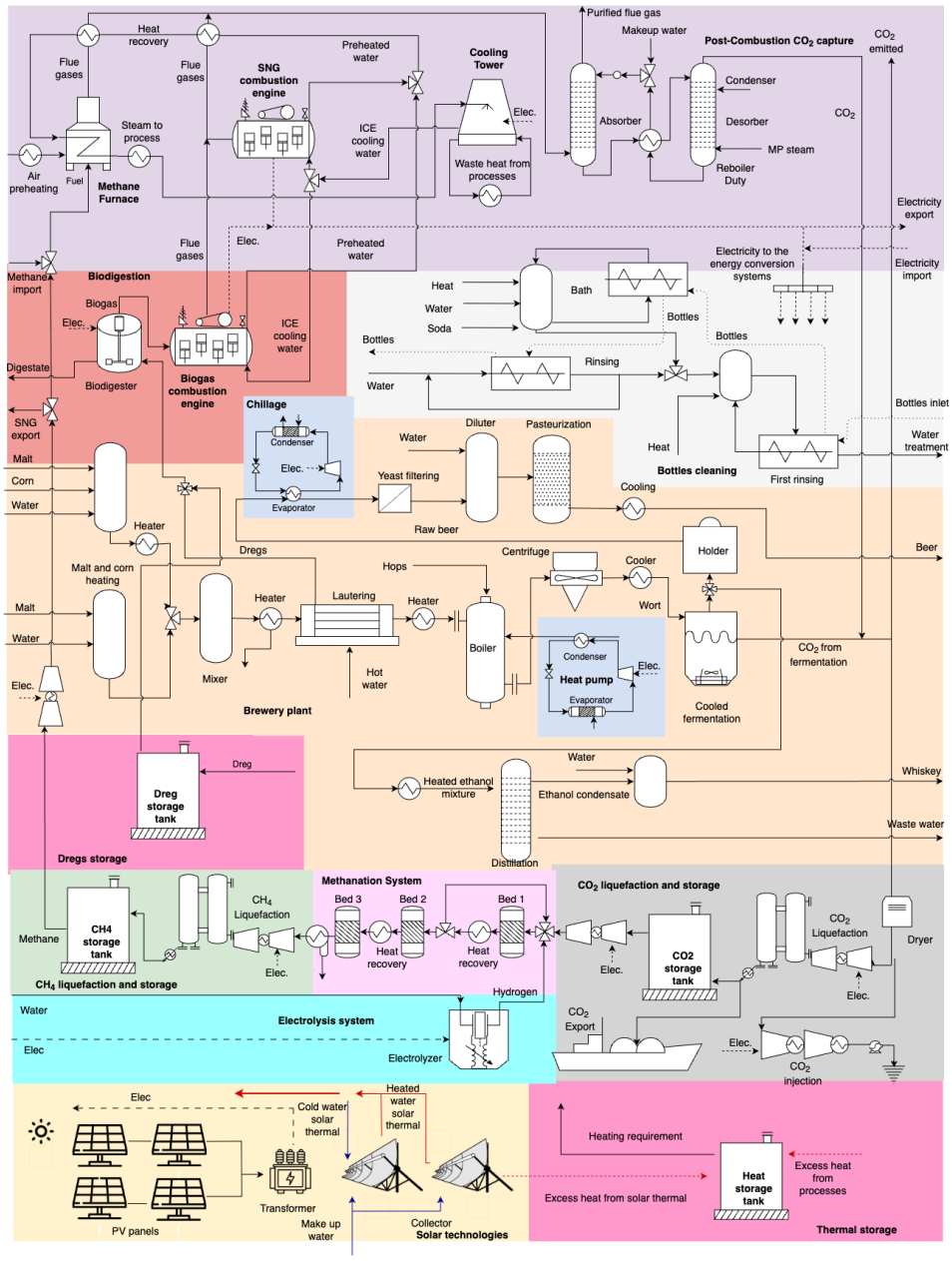


Figure 2: Superstructure of Brewery

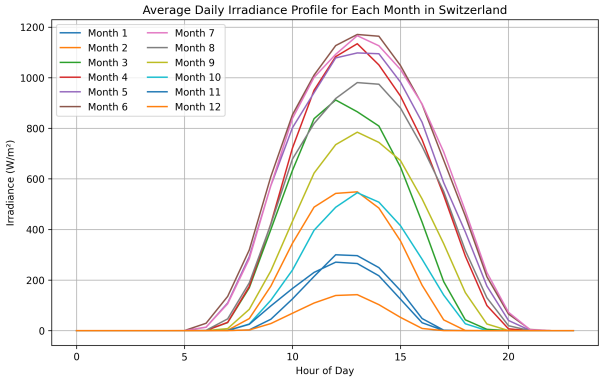


Figure 3: Average irradiance for a day in Switzerland

of optimizer (like Total Cost, Capital Cost and Operational Cost), while satisfying the thermodynamic constraints imposed by the MER problem [21].

AMPL CPLEX solver is utilized to solve this MILP problem. In this formulation, the integer variables, $y_{\omega,p}$, represent the existence or absence of any utility unit ω in the time period p, while corresponding continuous load factors $f_{\omega,p}$ denote their operating levels in a specific time period. The optimization aims to determine these variables by minimizing the objective function given by the Eq (1).

$$\min_{f_{\omega,p},y_{\omega,p},R_r,W} \left[\sum_{\omega=1}^{N_\omega} \max_{p\in\{1,\ldots,P\}} (f_{\omega,p}) * C_\omega + \sum_{\omega=1}^{N_\omega} \sum_{p=1}^{P} f_{\omega,p} * W_{\omega,p} * c_{\omega,p} * t_{OP} \right] \quad (1)$$

$$\sum_{\omega=1}^{N_\omega} f_{\omega,p} q_{\omega,r,p} + \sum_{i=1}^{N} Q_{i,r,p} + R_{r+1,p} - R_{r,p} = 0 \quad \forall r = 1,\ldots,N \quad \forall p = 1,\ldots,P \quad (2)$$

$$\sum_{\omega=1}^{N_\omega} f_{\omega,p} W_{\omega,p} + \sum_{\text{chemical units}} W_{\text{net,p}} + W_{\text{imp,p}} - W_{\text{exp,p}} = 0 \quad (3)$$

$$f_{\min,\omega,p}\, y_{\omega,p} \le f_{\omega,p} \le f_{\max,\omega,p}\, y_{\omega,p} \quad \forall \omega = 1,\ldots,N_\omega \quad \forall p = 1,\ldots,P \quad (4)$$

$$\sum_{\omega=1}^{N_\omega} y_{\omega,p} \leq 1, \quad y_{\omega,p} \in \{0,1\}, \quad f_{\omega,p} \geq 0 \quad \forall \omega = 1,\ldots,N_\omega \quad \forall p = 1,\ldots,P \tag{5}$$

$$R_{1,p} = 0, \quad R_{N+1,p} = 0, \quad R_{r,p} \geq 0, \quad W_{\text{imp}} \leq 0, \quad W_{\text{exp}} \geq 0 \tag{6}$$

$$C_\omega = C_0 \cdot \frac{r(1+r)^l}{(1+r)^l - 1} \tag{7}$$

Where, N represents the number of temperature intervals, determined by the supply and target temperatures of the process streams, P is the total number of periods a problem can be divided . Q denotes the heat exchanged between the process streams, where $Q_{i,r,p} > 0$ corresponds to a hot stream releasing heat in time period p and $Q_{i,r,p} < 0$ to a cold stream absorbing heat in a certain time period p. R refers to the heat flow rate (kW) that is cascaded from higher $(r+1)$ to lower (r) temperature intervals. The parameter N_ω indicates the total number of utility units considered in the system, while $\dot{m}$ represents the mass flow rate ($\mathrm{kg\,h^{-1}}$). In addition, q denotes the cooling or heating flow rates supplied by the utility systems (kW), and W is the power produced by the utility systems, chemical processes, or exchanged with the grid (kW). The parameter C_ω represents the annualized price (€/yr) of the utility unit integrated. The equation for the annualization of the cost is Eq(7), where Co is the initial capital cost, r is the interest rate and l is the life time of technology. c represents the price of carbon tax (€/kg), electricity consumption (€/kWh), or revenues from marketable products and byproducts. Finally, t_{OP} is the operation time (h/yr).

The conversion of organic material is modelled in DWSIM, while the auxiliary subsystems electrolyzer, carbon storage, methanation, and other utilities shown in Figure 2, are formulated as equation-oriented modules using Lua within the OSMOSE platform [[20]. Each module embeds the corresponding mass and energy balance relations following the multi-layer integration approach adopted in OSMOSE [[20], [22]], as illustrated in Figure 4 . In this framework, process units and utility systems interact simultaneously through dedicated layers representing heat, electricity, and material flows. The framework enables the consistent evaluation of energy and resource exchanges across the system. For example, the electricity balance in the power

layer ensures that the demand of process and auxiliary units is satisfied either through on-site generation (e.g., Dregs valorization) or grid exchange when required.

As shown in Figure 3, the solar irradiance profile varies significantly across months and within a typical day. To account for this temporal variability, the year is discretized into 12 representative months, each further divided into two characteristic days. This results in a total of 24 representative operating points, which are used to capture the variability in solar resource availability within the analysis. This temporal resolution is applied consistently across all configurations to ensure a fair comparison of technology performance.

The price variability is also a factor that influences the activation of storage.Following Gabrielli et al. [23], electricity price dynamics can be represented using a Fourier-based formulation. In this study, a simplified harmonic approximation is adopted, as expressed in Equation 8, to represent seasonal and short-term (intraday) price fluctuations.Here, CE_p denotes the electricity cost at time period p, CE_{avg} represents the average electricity price, CE_{vars} captures the amplitude of seasonal variation, and CE_{vard} accounts for intraday price deviations and where $I(day)$ is -1 for day and +1 for the night. In the current scenario, as described in above paragraph, the year is divided into 12 representative months, and each month is further divided into two days. So the equations give 24 different characteristic price

$$CE_{p,day} = CE_{\text{avg}} + CE_{\text{vars}} \cos\left(\frac{2\pi p}{12}\right) + CE_{\text{vard}}.I(day) \tag{8}$$

3.4. System-wide environmental analysis

In the current analysis LCA is performed from cradle to gate for all the three cases. Figure 5 illustrates the methodology followed for the estimating environmental impact of the conversion processes. The system is modeled using ecoinvent v3.10, while the LCIA is performed using Brightway, an open-source Python framework for impact calculation based on selected characterization methods. In this study, the IMPACT World+ v2.0.1 (footprint version) method is employed due to its global coverage and its ability to capture multiple environmental impact categories. The following five impact categories were used to assess the environmental emissions associated with the capture system.

- **CC**: Climate change, characterized by carbon footprint (kg CO_2eq)

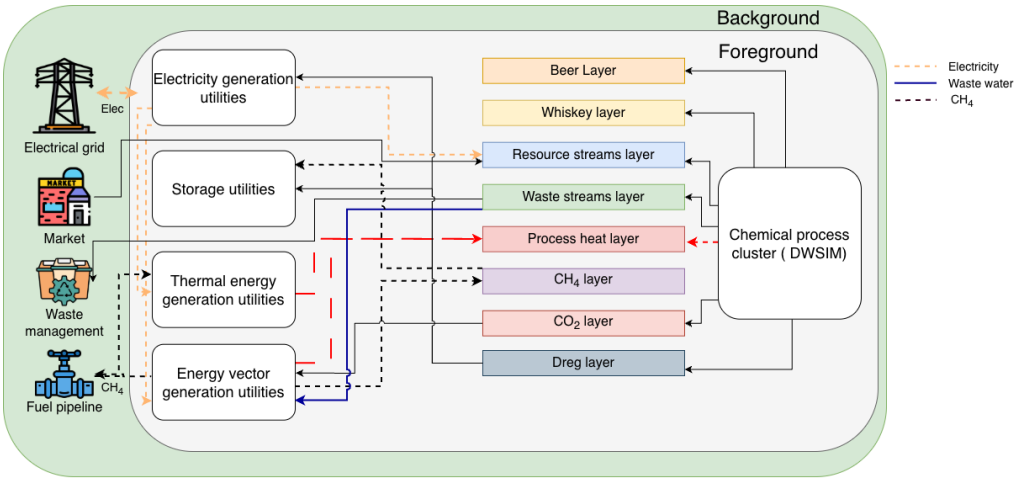


Figure 4: Multi-layer integration framework implemented in the OSMOSE platform for the conversion system.

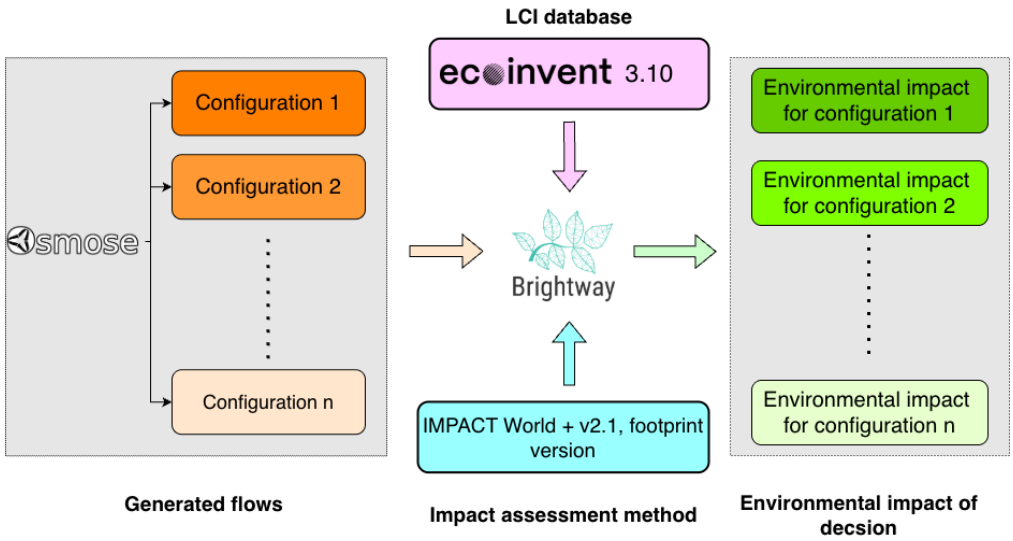


Figure 5: Creating a link between generated configuration and environmental impact of the configuration

- **EQ**: Ecosystem quality, measured in PDF·$\mathbf{m^2}$·yr (Potentially Disappeared Fraction of species over area and time)
- **NR**: Non-renewable energy use, quantified as MJ eq of fossil and nuclear energy
- **HH**: Human health, reported in DALYs (Disability-Adjusted Life Years) indicating potential harm to human well-being
- **WF**: Water use, assessed via the water scarcity footprint ($\mathbf{m^3}$ world eq)

These indicators capture both direct emissions and indirect resource burdens, which are particularly relevant for the conversion processes comparing the configuration .Background processes (e.g., electricity mix, transport, and infrastructure) are modeled using Swiss-specific data unless stated otherwise.

4. Results and discussion

This section presents and discusses the technical, economic, and environmental performance of the evaluated energy-system configurations for beer production, whiskey production, and combined beer–whiskey production. The analysis first quantifies opportunities for internal heat recovery, then evaluates alternative utility-integration configurations and their optimal selection under different economic and policy conditions. Finally, storage deployment and life-cycle environmental trade-offs are assessed.

4.1. Optimizing process heat distribution

Figure 6 shows the identified hot and cold streams for beer and whiskey . A sensitivity analysis on $\Delta T_{\min}$ (Table 1) was performed, and a value of 4 K was selected as a trade-off between energy efficiency and heat exchanger feasibility. Figure 7 presents the GCC, highlighting heat surpluses, deficits, and pinch points, based on a processing capacity of 10.26 ton/h of crops. For the combined case heat exchanging streams and GCC are illustrated in (Figure 8). It further implies the opportunities for cross-process heat integration.

The results demonstrate that heat integration reduces external heating demand by approximately 50–75%, depending on the process configuration. Beer production exhibits the highest heat recovery potential due to

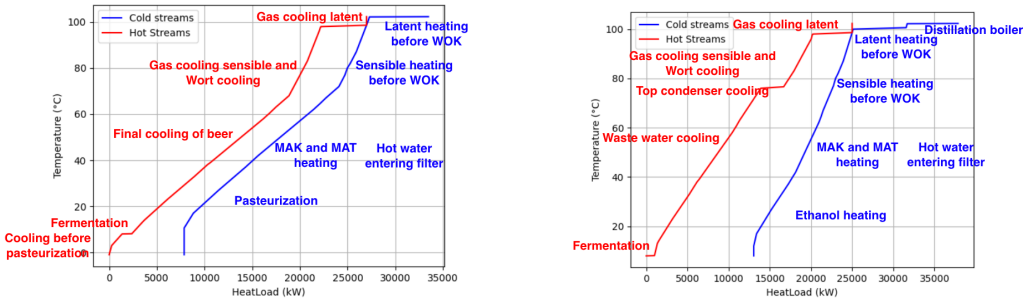


Figure 6: Hot and cold streams for beer (left) and whiskey (right) production processes

Table 1: Heating and cooling requirements under different heat integration scenarios

Product	No HEN		$\Delta T_{\min} = 2$		$\Delta T_{\min} = 4$		$\Delta T_{\min} = 6$	
	Heat (MWh/ton)	Cool (MWh/ton)	Heat (MWh/ton)	Cool (MWh/ton)	Heat (MWh/ton)	Cool (MWh/ton)	Heat (MWh/ton)	Cool (MWh/ton)
Beer	2.5	2.6	0.61	0.75	0.63	0.77	0.65	0.78
Whiskey	2.5	2.5	0.78	0.79	1.25	1.27	1.27	1.29
Combined	2.5	2.55	0.61	0.69	0.94	1.02	0.96	1.04

stronger overlap between hot and cold process streams and lower reliance on high-temperature separation duties. In contrast, whiskey production is constrained by distillation-related thermal requirements, which limits heat recovery and increases sensitivity to $\Delta T_{\min}$. A persistent pinch temperature of approximately 98.6°C is observed across all cases, indicating a common high-temperature thermodynamic bottleneck that cannot be resolved through internal heat integration alone and therefore defines the residual external utility demand addressed in the subsequent sections.

4.2. Utilities integration

Six utility-integration configurations are evaluated for beer production, whiskey production, and combined beer–whiskey production. The baseline configuration uses a natural-gas furnace for heating, a cooling tower for cooling above ambient temperature, and a refrigeration system for subambient cooling. The alternative configurations integrate heat pumps (HP), solar thermal energy with natural-gas backup (ST + NG), dregs valorization using a biogas engine combined with heat pumps (BGE + HP), bio-

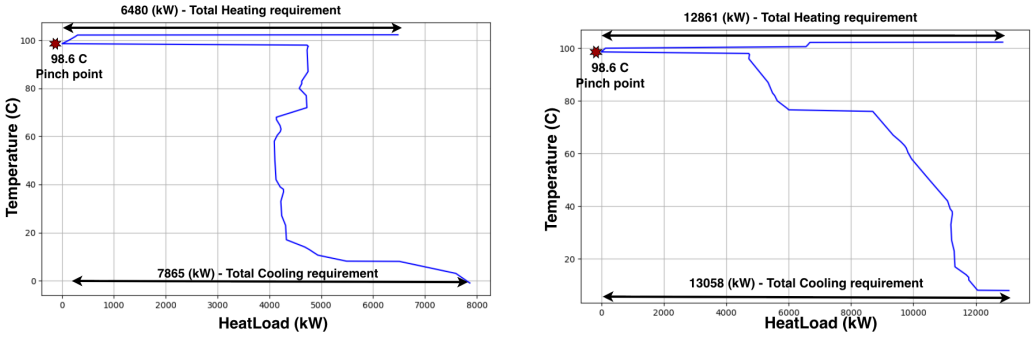


Figure 7: GCC for beer (left) and whiskey (right) production processes

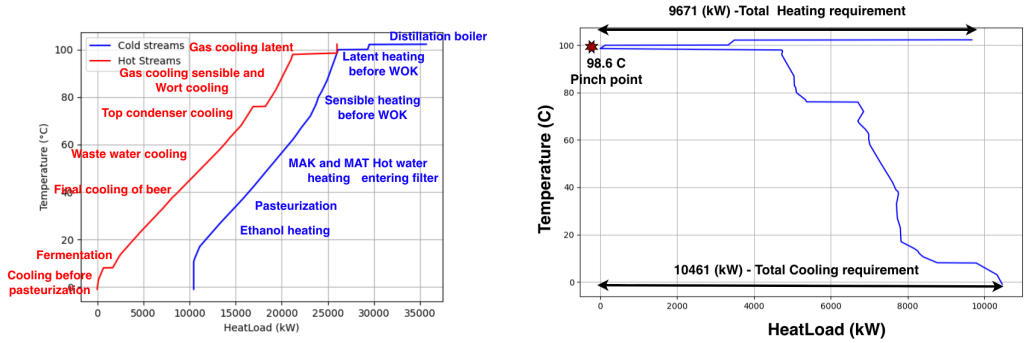
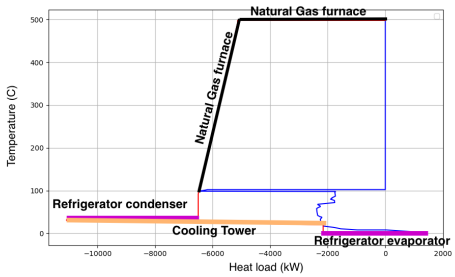


Figure 8: Hot and cold streams for the combined production process (left) and corresponding GCC (right)

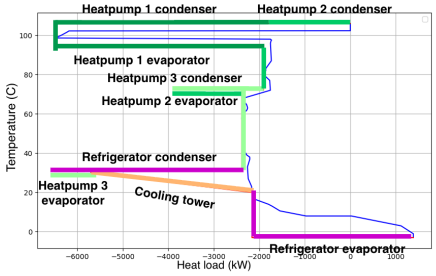
genic CO_2 methanation, and wastewater valorization with natural-gas backup (WWT + NG). Fossil CO_2 emissions associated with natural-gas and electricity consumption are calculated using emission factors of 0.243 and 0.100 tCO_2/MWh, respectively. The investment and annualization assumptions used to calculate the technology costs are reported in Table B.11

Table 2 compares utility consumption, CO_2 flows, synthetic methane production, and annualized capital costs across the evaluated configurations. In baseline scenario natural gas accounts for approximately 82%, 93%, and 89% of the combined natural-gas and electricity inputs for beer, whiskey, and combined production, respectively. Cooling demands are split relative to ambient temperature: cooling above ambient is provided by the cooling tower, while sub-ambient cooling is achieved using a refrigeration system. The refrigeration cycle extracts heat at low temperatures and rejects it at approximately 35°C, which is subsequently dissipated through the cooling tower. While the

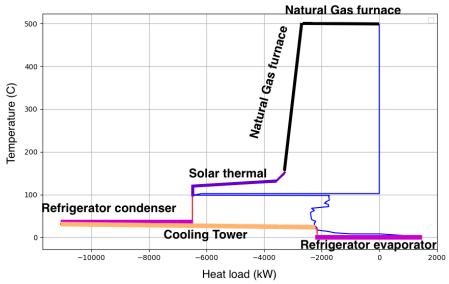
overall UGCC structure remains consistent across all cases, key differences arise from variations in refrigeration demand as shown in figures 9a–11a. In particular, the beer production case exhibits higher low-temperature cooling requirements, leading to increased electricity consumption for refrigeration.



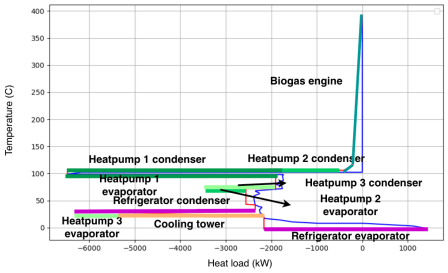
(a) Baseline or WWT +NG



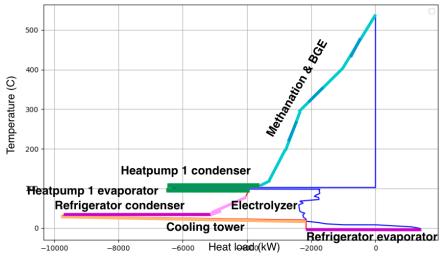
(b) HP



(c) ST + NG



(d) BGE + HP



(e) Methanation + HP

Figure 9: UGCC for the different utility options in the beer production case.

Table 2: Utility consumption, CO_2 emissions, synthetic methane production, and annualized capital cost for the evaluated utility-integration configurations.

(a) Beer production

Technology	Natural gas consumption MWh/t_{crop}	Electricity consumption MWh/t_{crop}	Fossil CO_2	Biogenic CO_2 released or utilized t_{CO_2}/t_{crop}	Facility-level net CO_2 balance	Synthetic CH_4 production MWh/t_{crop}	Annualized capital cost k€ yr^{-1}/(t_{crop} h^{-1})
Baseline	0.683	0.146	0.180	0.31	0.180	0.0	0.0
HP	0	0.280	0.028	0.31	0.028	0.0	58.1
ST + NG	0.574	0.146	0.150	0.31	0.150	0.0	70.7
BGE + HP	0	0.190	0.020	0.34	0.020	0.0	62.8
Methanation + HP	0	2.700	0.270	0.34	-0.070	1.5	275.3
WWT + NG	0.675	0.146	0.180	0.31	0.180	0.0	36.4

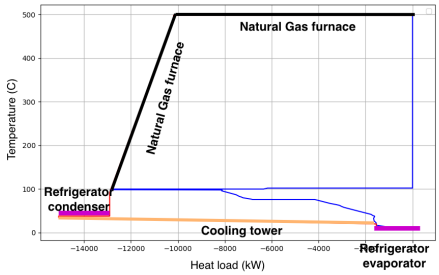
(b) Whiskey production

Technology	Natural gas consumption MWh/t_{crop}	Electricity consumption MWh/t_{crop}	Fossil CO_2	Biogenic CO_2 released or utilized t_{CO_2}/t_{crop}	Facility-level net CO_2 balance	Synthetic CH_4 production MWh/t_{crop}	Annualized capital cost k€ yr^{-1}/(t_{crop} h^{-1})
Baseline	1.340	0.104	0.340	0.34	0.340	0.0	0.0
HP	0	0.306	0.030	0.34	0.030	0.0	110.3
ST + NG	1.240	0.104	0.310	0.34	0.310	0.0	70.7
BGE + HP	0	0.250	0.025	0.37	0.025	0.0	115.0
Methanation + HP	0	3.100	0.300	0.37	-0.070	1.7	356.4
WWT + NG	1.320	0.104	0.330	0.34	0.330	0.0	78.5

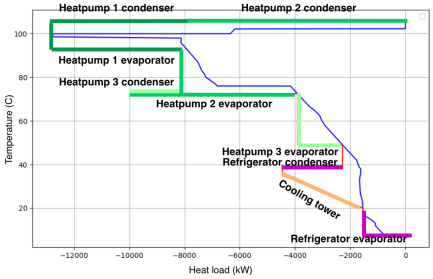
(c) Combined beer and whiskey production

Technology	Natural gas consumption MWh/t_{crop}	Electricity consumption MWh/t_{crop}	Fossil CO_2	Biogenic CO_2 released or utilized t_{CO_2}/t_{crop}	Facility-level net CO_2 balance	Synthetic CH_4 production MWh/t_{crop}	Annualized capital cost k€ yr^{-1}/(t_{crop} h^{-1})
Baseline	1.010	0.125	0.260	0.33	0.260	0.0	0.0
HP	0	0.290	0.029	0.33	0.029	0.0	84.1
ST + NG	0.900	0.125	0.230	0.33	0.230	0.0	70.7
BGE + HP	0	0.220	0.022	0.36	0.022	0.0	88.8
Methanation + HP	0	2.900	0.290	0.36	-0.070	1.6	314.9
WWT + NG	1.000	0.125	0.260	0.33	0.260	0.0	57.4

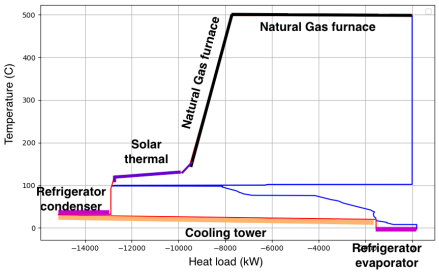
Note: Fossil CO_2 emissions are calculated from natural-gas and electricity consumption using emission factors of 0.243 and 0.100 tCO_2/MWh, respectively. The facility-level CO_2 balance equals fossil CO_2 emissions for configurations without methanation. For the Methanation + HP configuration, the balance is calculated by subtracting the biogenic CO_2 transferred into synthetic methane from the fossil CO_2 emissions. The system boundary terminates when beer, whiskey, and synthetic methane leave the production facility. The negative balance does not represent permanent carbon removal because the downstream use of synthetic methane is outside the system boundary.



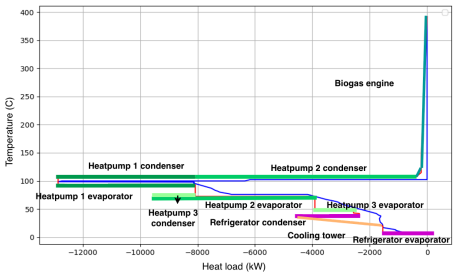
(a) Baseline or WWT +NG



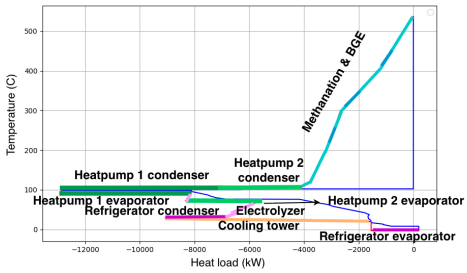
(b) HP



(c) ST + NG

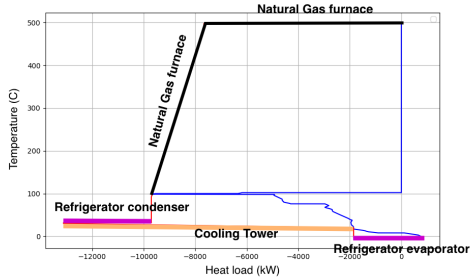


(d) BGE + HP

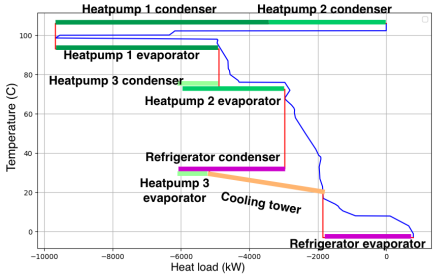


(e) Methanation + HP

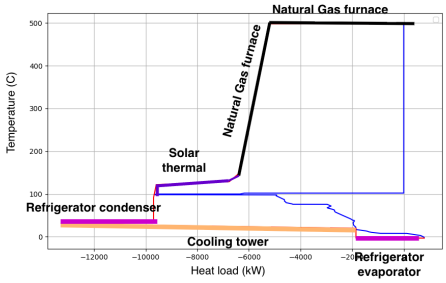
Figure 10: UGCC for the different utility options in the whiskey production case.



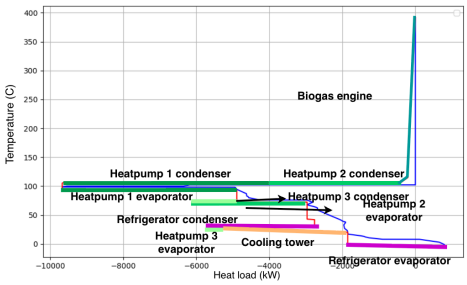
(a) Baseline or WWT +NG



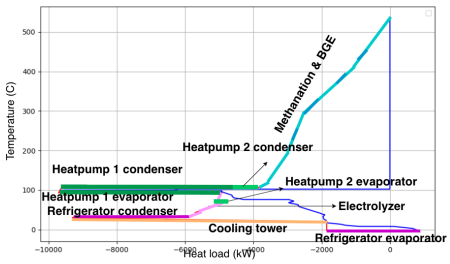
(b) HP



(c) ST + NG



(d) BGE + HP



(e) Methanation + HP

Figure 11: UGCC for the different utility options in the combined production case.

Heat-pump integration

Heat-pump integration reduces external final-energy input by approximately 66%, 79%, and 75% relative to the corresponding baseline configurations for beer, whiskey, and combined production, respectively. Fossil CO_2 emissions are reduced by approximately 84–91%, although these reductions require additional annualized capital investment. The improvements arise because the heat pumps recover heat from process streams requiring cooling

and upgrade it to temperature levels at which it can satisfy process heating demands. Figures 9b–11b illustrate the integration of the heat pumps at different temperature levels. Three heat pumps are considered in each production scenario to account for the different temperature levels of the available heat sources and process heat demands. The corresponding condenser temperatures, evaporator temperatures, and coefficients of performance (COPs) are reported in Table 3. The heat-pump operating temperatures are selected to reduce thermodynamic losses while maximizing the recovery of low-grade heat below the pinch. In all three production scenarios, a significant heat source is available at approximately 98 °C. However, the amount of heat available at this temperature is insufficient to satisfy the entire heating demand above the pinch.

For beer production, the remaining heating demand is satisfied by upgrading heat recovered from the condenser of the refrigeration system. For whiskey and combined production, an additional heat source is available at approximately 76 °C. Consequently, an evaporator temperature of approximately 70.5 °C is selected for the second heat pump. This intermediate-temperature source remains insufficient to satisfy the complete heating demand. In the whiskey-production case, the remaining demand is met using an additional heat source available at approximately 45 °C. In the combined-production case, residual heat demand is met by recovering and upgrading heat from the refrigeration condenser.

The emission reductions achieved through heat-pump integration are accompanied by additional equipment investment. Table 4 reports the normalized installed capacity and annualized capital cost of each heat pump. The whiskey-production case requires the largest total heat-pump capacity, primarily because of the greater heating demand associated with the distillation process.

Table 3: Operating temperatures and coefficients of performance of the heat pumps selected for the three production scenarios.

	HP 1			HP 2			HP 3		
Product	Cond(°C)	Evap(°C)	COP	Cond(°C)	Evap(°C)	COP	Cond(°C)	Evap(°C)	COP
Beer	108	93	12.70	108	70	5.01	75	30	3.94
Whiskey	108	91	11.21	108	70.5	5.08	75.5	45	5.89
Combined	108	91	11.21	108	70.5	5.08	75.5	30	3.91

Table 4: Normalized installed capacities and annualized capital costs of the heat pumps selected for the three production scenarios.

Product	HP 1		HP 2		HP 3	
	Size $kW/(t_{crop}\,h^{-1})$	CAPEX $k€\,yr^{-1}/(t_{crop}\,h^{-1})$	Size $kW/(t_{crop}\,h^{-1})$	CAPEX $k€\,yr^{-1}/(t_{crop}\,h^{-1})$	Size $kW/(t_{crop}\,h^{-1})$	CAPEX $k€\,yr^{-1}/(t_{crop}\,h^{-1})$
Beer	448	34.2	147.5	11.3	165.1	12.6
Whiskey	511.7	39.1	741.9	56.7	189.3	14.5
Combined	511.7	39.1	430.9	32.9	158.5	12.1

Solar thermal integration

In the ST + NG configuration, solar thermal energy is used to satisfy part of the process heating demand, while the residual demand is supplied by the natural-gas furnace. Relative to the baseline configurations, solar thermal integration reduces external final-energy input by approximately 7–13% and fossil CO_2 emissions by approximately 9–17%. The available solar collector area is limited to approximately 15,000 m^2. Figures 9c–11c presents the UGCC for July, which represents the period of peak solar availability. Even under these favorable conditions, a residual heating demand remains above the pinch. This residual demand results from both the limited amount of available solar heat and the mismatch between the temperature of the solar heat supply and the temperature of the process demand above the pinch. Natural gas therefore remains necessary to satisfy the remaining high-temperature heating requirements. Figure 12 presents the temporal contribution of solar thermal energy across the representative operating periods. The contribution varies according to seasonal solar availability, whereas the process heating demand must be satisfied throughout the year.

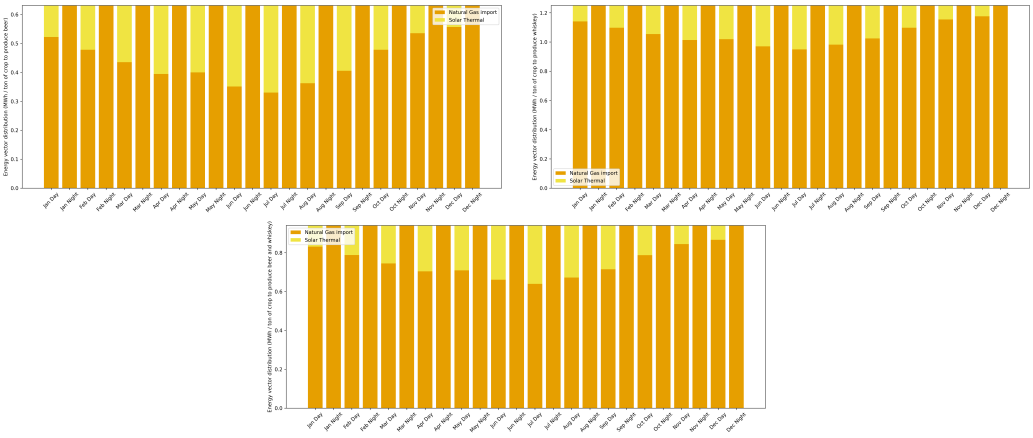


Figure 12: Contribution of solar thermal energy to process heating demand across the representative operating periods.

Dregs valorization and biogas-engine integration

Combining dregs valorization with heat pumps results in the lowest external energy consumption and fossil CO_2 emissions among the configurations that do not export synthetic methane. Fossil CO_2 emissions are reduced by approximately 89%, 93%, and 92% relative to the baseline configurations for beer, whiskey, and combined production, respectively. Figures 9d – 11d illustrate the integration of the biogas engine and heat pumps. The recovered energy reduces dependence on external energy carriers. However, anaerobic digestion and biogas combustion generate additional biogenic CO_2, increasing the available biogenic CO_2 from 0.31 to 0.34, 0.34 to 0.37, and 0.33 to 0.36 tCO_2/t_{crop} for beer, whiskey, and combined production, respectively.

Biogenic CO_2 methanation

The methanation configuration has the highest external electricity consumption, primarily because of the electricity required to produce hydrogen through water electrolysis. Under the assumed electricity emission factor, it consequently has the highest fossil CO_2 emissions among the electrified configurations. Nevertheless, the configuration converts the available biogenic CO_2 into 1.5–1.7 MWh of synthetic methane per tonne of crop using biogenic CO_2 as the carbon source. A portion of heating demand is supplied using heat pumps and BGE as shown in Figures 9e–11e. Methanation is also the most capital-intensive configuration because it requires an electrolyzer, methanation reactor, heat pumps, and a biogas engine. Table 5 reports the normalized

equipment capacities and annualized capital costs. Within the facility-level carbon balance adopted in this study, the biogenic CO_2 transferred into synthetic methane exceeds the fossil CO_2 emissions associated with external electricity consumption, resulting in a balance of -0.07 tCO_2/t_{crop} for all three production scenarios. This value does not represent permanent atmospheric carbon removal because the synthetic methane leaves the facility as a co-product and its downstream use lies outside the system boundary.

Table 5: Normalized equipment capacities and annualized capital costs for the configuration converting biogenic CO_2 into synthetic methane.

Product	Electrolyzer		Methanator		HP 1		HP 2		BGE	
	Size	CAPEX	Size	CAPEX	Size	CAPEX	Size	CAPEX	Size	CAPEX
	$kW/(t_{crop}\,h^{-1})$	$k€\,yr^{-1}/(t_{crop}\,h^{-1})$	$kW/(t_{crop}\,h^{-1})$	$k€\,yr^{-1}/(t_{crop}\,h^{-1})$	$kW/(t_{crop}\,h^{-1})$	$k€\,yr^{-1}/(t_{crop}\,h^{-1})$	$kW/(t_{crop}\,h^{-1})$	$k€\,yr^{-1}/(t_{crop}\,h^{-1})$	$kW/(t_{crop}\,h^{-1})$	$k€\,yr^{-1}/(t_{crop}\,h^{-1})$
Beer	2558	223	1291	28.1	271	20.7	0	0	41	3.5
Whiskey	2937	256	1483	32.3	511.7	39.1	334	25.5	41	3.5
Combined	2747	239.5	1388	30.2	511.7	39.1	47	3.6	41	3.5

Wastewater valorization

The amount of wastewater generated differs among the production scenarios. Whiskey production generates approximately 8.2 m^3/t_{crop}, compared with 3.8 m^3/t_{crop} for beer production. The greater wastewater generation in whiskey production primarily results from the distillation process, which produces substantial quantities of liquid stillage.

The larger wastewater flow in whiskey production provides a greater theoretical potential for biogas production. However, the methane yields obtained under the assumed wastewater characteristics are insufficient to replace a substantial fraction of natural-gas consumption. Consequently, WWT + NG reduces fossil CO_2 emissions by only approximately 0–3% relative to the baseline configurations, while requiring an annualized capital investment of 36.4–78.5 k€/yr. Under the assumptions adopted in this study, wastewater valorization therefore provides limited energy and emission benefits relative to its capital cost.

4.3. Selection of technologies through MILP optimization

The preceding subsection evaluates the technical performance of representative utility-integration configurations. The present subsection extends this analysis by allowing the MILP model to select and size technologies from the complete superstructure. Consequently, the optimal solutions are not restricted to the predefined configurations discussed previously and may include hybrid combinations of natural gas, solar thermal energy, heat pumps, waste valorization, methanation, and storage.

The available technologies can be broadly classified into two categories. First, *cost-driven technologies* are selected based on their ability to meet process energy demands at minimum total cost under given utility prices. These include heat pumps, solar thermal systems, natural gas furnaces, dregs valorization through biogas engines, wastewater treatment, and biogenic CO_2 utilization pathways. Second, *storage technologies* provide operational flexibility by mitigating temporal mismatches between energy supply and demand, particularly under fluctuating electricity prices and intermittent renewable availability. These include biomass (dregs) storage, thermal energy storage, and CO_2 and CH_4 storage systems.

The system is optimized with respect to TOTEX, such that the resulting configurations reflect a trade-off between capital-intensive technologies, operational efficiency, and flexibility under different boundary conditions.

4.3.1. Impact of energy prices and carbon tax on technology selection

Figures 13–15 illustrate the optimal technology configurations across a wide range of electricity prices, natural gas prices, and carbon tax levels for the three production cases. The results reveal distinct operational regimes governed by the relative cost of energy carriers and carbon constraints.

At low electricity and natural gas prices (below 0.1 €/kWh), the system is consistently dominated by heat pump-based configurations. This reflects the high COP of heat pumps, which enables efficient upgrading of low-temperature heat while simultaneously reducing both heating and cooling utility requirements. Notably, heat pumps remain part of the optimal solution even at elevated electricity prices (up to 0.5 €/kWh), particularly for high-performance units (e.g., HP1). This persistence highlights their systemic benefit: reductions in cooling demand partially offset increased electricity costs, thereby maintaining overall cost competitiveness.

As electricity and natural gas prices increase, a transition toward renewable heat supply is observed, with ST systems becoming economically attractive and often deployed alongside HP. These hybrid configurations reflect a trade-off between higher capital costs and lower operating costs, reducing reliance on expensive utilities. At the upper end of the evaluated price range, particularly when the natural-gas price approaches 0.3 €/kWh and the electricity price approaches 0.5 €/kWh, HP-dominated configurations again become optimal, and ST is no longer deployed. This is because ST alone cannot fully meet heat demand and requires natural gas backup; when gas prices are high, this hybrid becomes economically unattractive, favoring

standalone heat pump systems.

Carbon pricing shifts the optimal solution away from natural-gas-based heating by increasing its effective operating cost. As the carbon tax increases, natural-gas furnace capacity is progressively displaced by heat pumps and solar thermal systems. Methanation becomes attractive only when the carbon-price incentive is accompanied by sufficiently low electricity prices, because hydrogen production through electrolysis is electricity intensive.

Under conditions of low electricity prices (below 0.05 €/kWh) and high natural gas prices (above 0.3 €/kWh), CO_2 valorization pathways, including methanation, become economically favorable. In these scenarios, the exothermic heat released during methanation contributes to process heat demand, while the produced methane provides an additional revenue stream. This dual benefit demonstrates the potential of carbon-utilization technologies to valorize biogenic CO and improve economic performance under favourable electricity-price and carbon-policy conditions. Their overall emissions performance nevertheless depends strongly on the carbon intensity of the electricity used for hydrogen production.

Four technology-selection regimes can be distinguished: (i) an HP-dominated regime, enabled by the high COP and simultaneous provision of heating and cooling; (ii) a hybrid ST + HP regime at intermediate-to-high utility prices, where solar heat reduces purchased-energy consumption; (iii) a return to HP-dominated configurations at the highest natural-gas prices, where the need for natural-gas backup reduces the attractiveness of solar thermal integration; and (iv) a methanation regime at low electricity and high natural-gas prices, particularly when carbon pricing improves the value of biogenic CO_2 utilization.

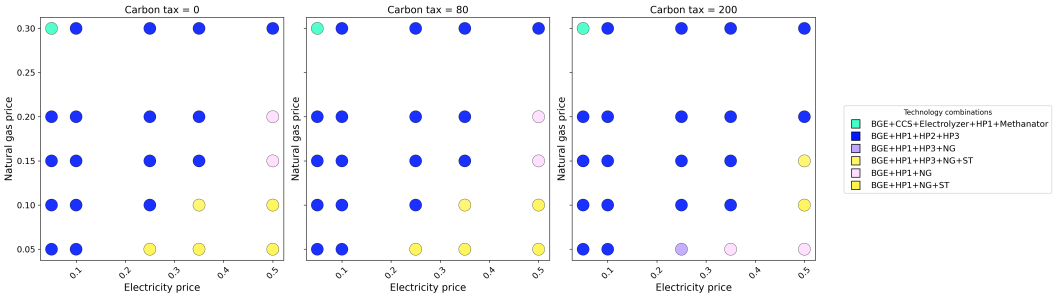


Figure 13: Activation of technologies for different scenarios during beer production

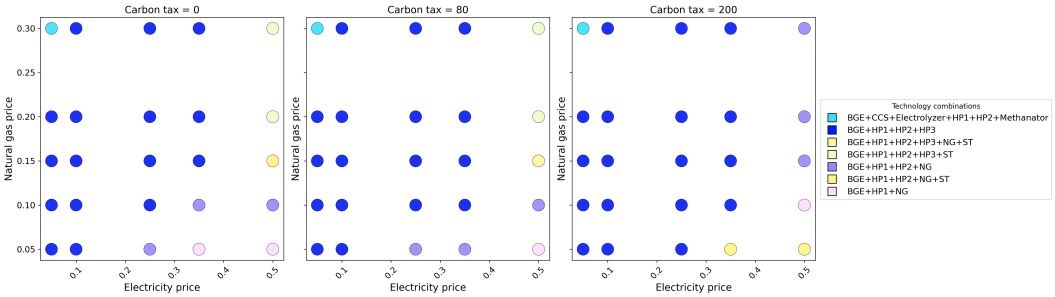


Figure 14: Activation of technologies for different scenarios during whiskey production

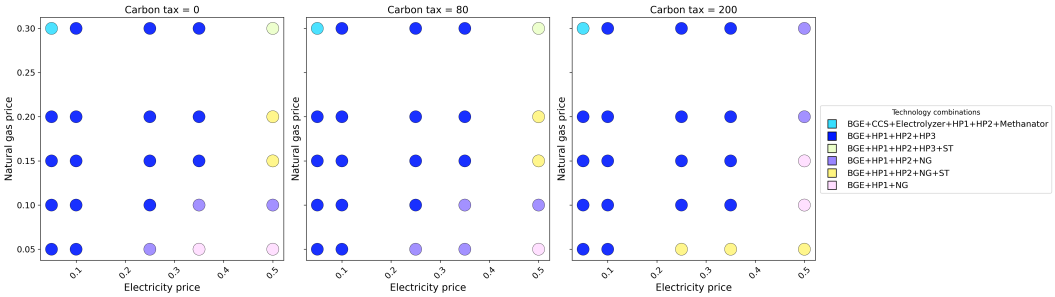


Figure 15: Activation of technologies for different scenarios during beer and whiskey production

4.3.2. Role of storage technologies

The preceding sensitivity analysis evaluates technology selection for fixed combinations of average electricity price, natural-gas price, and carbon tax. The present analysis extends this framework by introducing temporal electricity-price variability and inter-period material balances. Storage can therefore shift the use of internally available resources between operating periods and reduce exposure to periods of high energy prices.

Several storage options are included in the technological superstructure:dregs storage, thermal energy storage, CO_2 storage, and CH_4 storage. Among these options, only dregs storage is selected within the investigated economic conditions. Thermal energy storage is not selected because of the limited amount of recoverable heat above the pinch and the resulting limited opportunity for temporal heat shifting. CO_2 storage is not competitive because of the additional compression requirements and capital cost. Similarly, CH_4 storage does not generate sufficient temporal-arbitrage value to offset its ad-

ditional investment cost, particularly because synthetic methane production is already electricity and capital intensive.

Since dregs storage is the only storage option selected within the investigated economic parameter space, it is examined in greater detail. In particular, the analysis identifies how its economic viability and associated utility configuration depend on the natural-gas price, average electricity price, and electricity-price variability.

Dregs storage provides long-term flexibility by allowing the by-product to be stored during one operating period and converted into biogas during another. The stored dregs can subsequently be used in the biogas engine to generate heat and electricity when the economic value of internal energy production is higher. Its activation is therefore governed by the natural-gas price, the average electricity price (CE_{avg}), and the electricity-price variability parameter (CE_{vars}), as defined in Section **??**.

Figure 16 presents the regions in which dregs storage is activated for natural-gas prices of 0.3, 0.2, and 0.1 €/kWh. For a given value of CE_{avg}, storage is selected when CE_{vars} exceeds the activation boundary shown in each panel. The region above the boundary therefore represents the combinations of average electricity price and electricity-price variability for which the economic benefit of shifting dregs use between operating periods exceeds the additional storage cost.

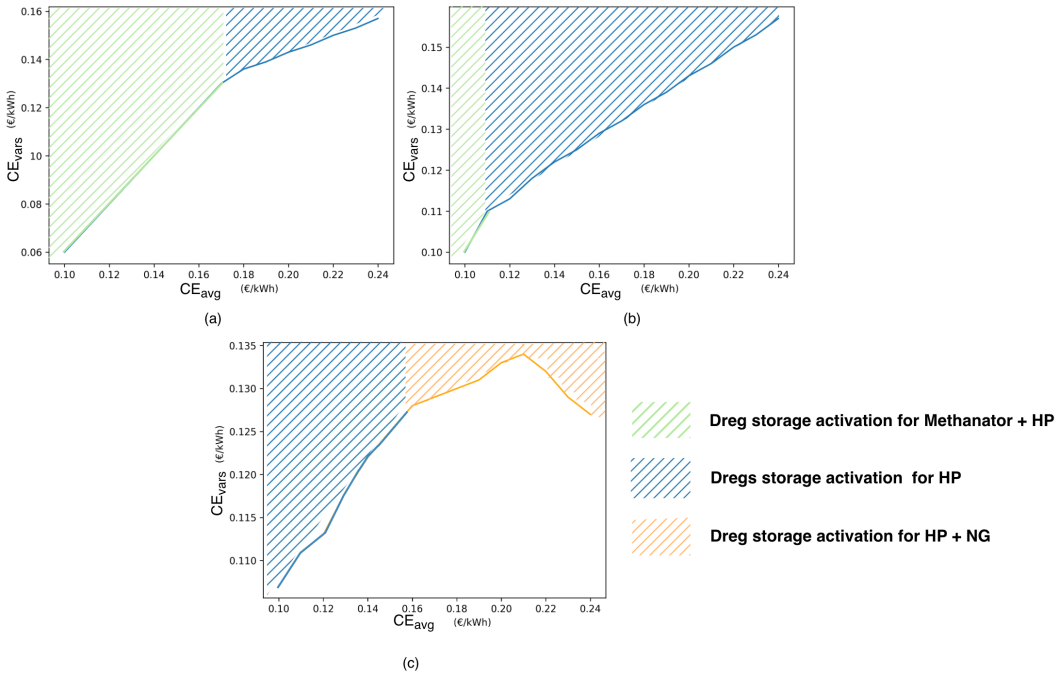


Figure 16: Activation regions of dregs storage as a function of average electricity price (CE_{avg}) and electricity-price variability (CE_{vars}) at natural-gas prices of:(a) 0.3 €/kWh, (b) 0.2 €/kWh, and (c) 0.1 €/kWh. The hatched regions indicate conditions under which dregs storage is selected, while the colours identify the associated optimal utility configuration.

At a natural-gas price of 0.3 €/kWh (Figure 16(a)), dregs storage is activated over a comparatively broad range of electricity-price conditions. The high natural-gas price increases the value of substituting externally purchased natural gas with internally produced biogas. At lower average electricity prices, storage activation is associated primarily with the Methanation + HP configuration. In this regime, dregs are preserved during periods in which methanation heat is available and are subsequently converted into biogas when their use for on-site heat and electricity generation provides a greater economic benefit. At higher average electricity prices, the HP configuration becomes dominant, and storage is used to reduce electricity purchases during high-price periods.

When the natural-gas price decreases to 0.2 €/kWh (Figure 16(b)), the economic value of substituting natural gas with dregs-derived biogas is reduced. Consequently, for a given average electricity price, a larger value of

CE_{vars} is generally required to activate storage. In this case, electricity-price arbitrage becomes the principal driver of storage deployment. The Methanation + HP activation region becomes considerably smaller, while storage associated with the HP configuration dominates most of the investigated price space.

At a natural-gas price of 0.1 €/kWh (Figure 16(c)), the incentive to store dregs for the purpose of natural-gas substitution is further weakened. The Methanation + HP configuration is no longer selected. At lower values of CE_{avg}, storage activation occurs within the HP configuration and requires a sufficiently strong electricity-price variability signal. At higher average electricity prices, an HP + NG configuration becomes optimal. In this region, dregs storage is activated when the value of displacing expensive electricity during selected operating periods is sufficient to recover the storage cost.

Overall, the comparison of Figures 16a–c shows that natural-gas price and electricity-price variability act as alternative economic drivers for dregs storage. At high natural-gas prices, the value of replacing natural gas with internally produced biogas can justify storage even under relatively limited electricity-price variability. As the natural-gas price decreases, this substitution benefit weakens, and a stronger electricity-price variability signal is generally required at the same average electricity price. The precise activation boundary is also influenced by changes in the optimal utility configuration, explaining the non-monotonic behavior observed in some regions of Figure 16c.

4.3.3. Representative optimal configurations

Tables 6–8 summarize these representative configurations for the three production cases.

Across all cases, several consistent patterns emerge. Heat pumps constitute the core technology in most configurations, reflecting their high efficiency and central role in reducing overall utility demand. Solar thermal systems are integrated in scenarios characterized by high energy prices or strong carbon constraints, while natural gas-based heating remains relevant primarily in low-cost regimes.

Biomass valorization through biogas engines is systematically present, highlighting the importance of internal resource utilization. In contrast, methanation-based configurations appear only under specific economic conditions, namely low electricity prices and high natural gas prices, where carbon utilization becomes economically advantageous.

Storage integration is strongly correlated with energy price variability and renewable intermittency. Among the evaluated storage options, dregs storage is the only technology selected within the broader sensitivity space. However, it is not activated in the representative configurations reported in Tables 6–8. These representative cases either require short-term thermal balancing, which dregs storage cannot provide, or exhibit insufficient electricity-price variability to recover the additional storage cost.

Table 6: Representative optimal technology configurations for beer production.

Configuration	NG kW	ST kW	HP1 kW	HP2 kW	HP3 kW	Methanator kW	Electrolyzer kW	BGE kW
1	680	0	0	0	0	0	0	0
2	0	0	484	105	148	0	0	41
3	187	0	416	0	0	0	0	41
4	113	0	484	0	64	0	0	41
5	113	117	484	0	63.57	0	0	41
6	112	147	484	0	66.20	0	0	41
7	0	0	271	0	0	1291	2558	41
8	112	122	484	0	63.87	0	0	41
9	106	151	484	0	66.83	0	0	41
10	187	193	415	0	0	0	0	41
11	109	140	484	0	65	0	0	41

Table 7: Representative optimal technology configurations for whiskey production.

Configuration	NG kW	ST kW	HP1 kW	HP2 kW	HP3 kW	Methanator kW	Electrolyzer kW	BGE kW
1	1340	0	0	0	0	0	0	0
2	0	0	512	700	156	0	0	41
3	208	0	512	507	0	0	0	41
4	753	0	512	0	0	0	0	41
5	204	240	512	507	0	0	0	41
6	0	63	512	680	140	0	0	41
7	0	0	511.7	334	0	1483	2937	41
8	174	233	512	531	19.4	0	0	41
9	203	241	512	507	0	0	0	41

4.4. Life Cycle Assessment analysis

A cradle-to-gate life cycle assessment was performed for each representative optimal configuration identified by the MILP model. The foreground

Table 8: Representative optimal technology configurations for combined beer and whiskey production.

Configuration	NG kW	ST kW	HP1 kW	HP2 kW	HP3 kW	Methanator kW	Electrolyzer kW	BGE kW
1	1010	0	0	0	0	0	0	0
2	0	0	512	389	125	0	0	41
3	166.5	0	512	234	0	0	0	41
4	418	0	512	0	0	0	0	41
5	162	202	512	234	0	0	0	41
6	156.5	226	512	234	0	0	0	41
7	0	0	511.7	47	0	1388	2747	41
8	0	74	512	366	106	0	0	41
9	159	214	512	234	0	0	0	41
10	162	202	512	234	0	0	0	41

system includes beer production, whiskey production, or combined production up to the facility gate, while upstream processes associated with crop cultivation, material supply, electricity generation, natural-gas supply, and technology operation are included through the background life cycle inventory.

4.4.1. Configuration-driven environmental trade-offs

The representative configurations identified by the MILP optimization are evaluated using LCA to determine whether reductions in operating cost and direct fossil CO_2 emissions are accompanied by reductions in broader life cycle impacts. Across all production systems, the results reveal pronounced variability in environmental performance across configurations, highlighting the strong dependence of life cycle impacts on technology selection. The configuration numbers used in the LCA correspond to those reported in Tables **??**–8. Table9 depicts the environmental impact across different impact categories for the baseline (configuration 1).

Table 9: Cradle-to-gate life cycle impacts of the baseline configurations per tonne of crop processed.

Impact category	Unit	Beer	Whiskey	Combined
Carbon footprint (CC)	kg CO_2-eq	567.4	658.4	613.0
Non-renewable energy use (NR)	MJ	6308	7580	6947
Water footprint (WF)	m^3 world-eq	1734	1392	1620
Human health (HH)	DALY	0.0110	0.0110	0.0114
Ecosystem quality (EQ)	PDF m^2 yr	58084.3	58118.8	58101.3

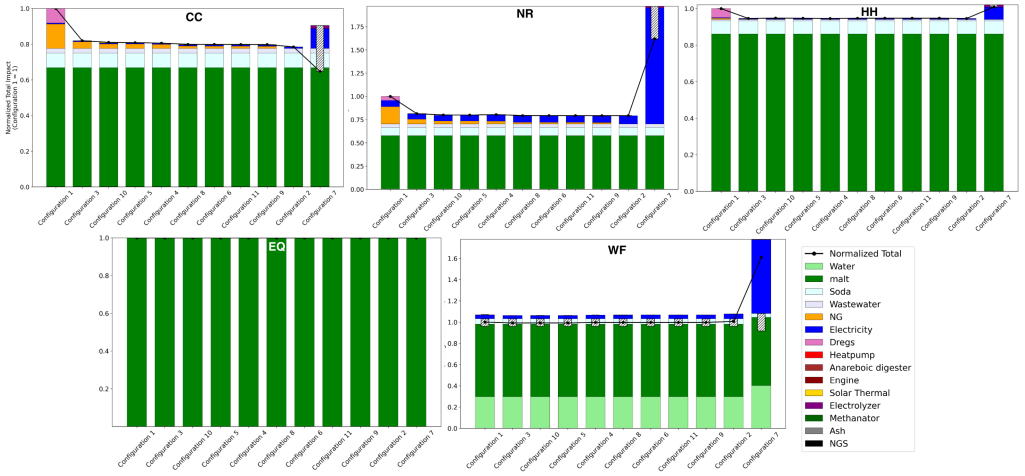


Figure 17: Normalized life cycle impacts and process contributions of the representative optimal configurations for beer production. Total impacts are normalized relative to the baseline configuration.

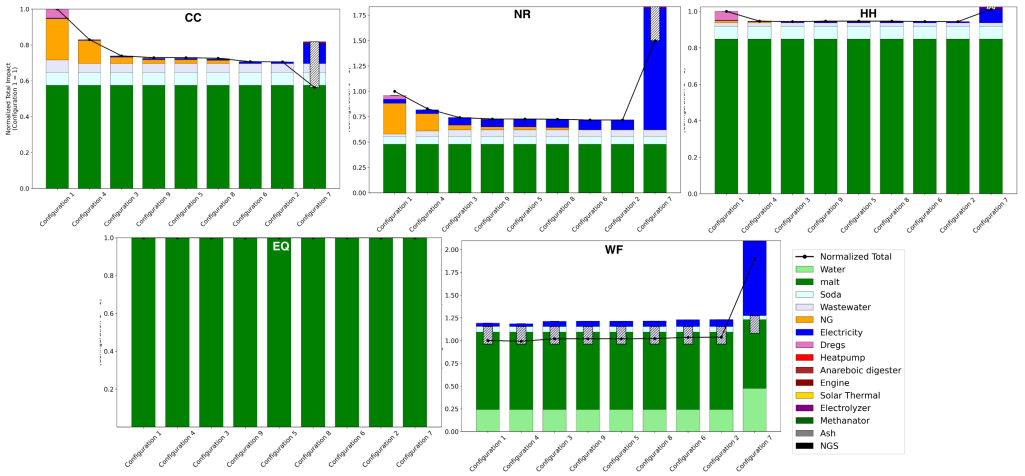


Figure 18: Normalized life cycle impacts and process contributions of the representative optimal configurations for whiskey production. Total impacts are normalized relative to the baseline configuration.

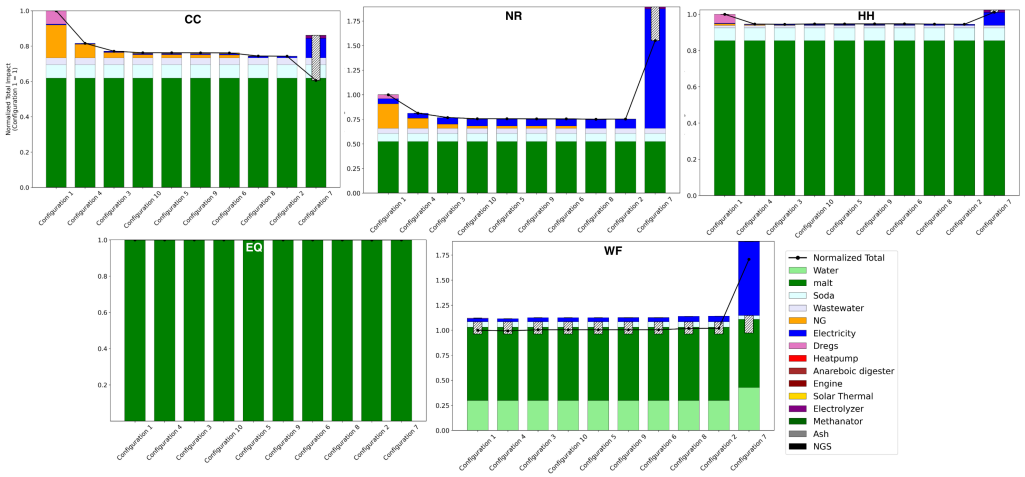


Figure 19: Normalized life cycle impacts and process contributions of the representative optimal configurations for combined beer and whiskey production. Total impacts are normalized relative to the baseline configuration.

Figures ??–?? compare the life cycle impacts of the representative optimal configurations for beer, whiskey, and combined production. The total impact of each configuration is normalized relative to the corresponding baseline configuration, while the stacked bars disaggregate the contributions of

individual processes, resources, and utility technologies.

Across the three production systems, all alternative configurations reduce the carbon footprint relative to the baseline. These reductions result primarily from lower natural-gas consumption and the integration of heat pumps, solar thermal systems, and internally generated biogas. The largest carbon-footprint reductions are generally achieved by configurations combining heat pumps with partial natural-gas substitution or waste valorization.

Heat-pump-dominated configurations also reduce non-renewable energy use and, in most cases, human-health impacts. Their environmental advantage results from the recovery and upgrading of heat below the pinch, which reduces the external energy required to satisfy both process heating and cooling demands. However, these configurations do not reduce all environmental indicators. Water-related impacts generally increase slightly because of the greater electricity consumption, while ecosystem-quality impacts remain nearly unchanged.

Ecosystem-quality impacts are dominated by malt production and vary only marginally among the evaluated configurations. This is because the amount of crop processed remains constant, irrespective of the selected utility system. Consequently, changes in utility technologies have a limited influence on agriculture-dominated environmental indicators. Contribution analysis further indicates that fertilizer production and application are important sources of the impacts associated with malt production.

Configuration 7, which integrates biogenic CO_2 methanation, exhibits a distinct burden-shifting pattern. Although its carbon footprint remains below that of the baseline, its non-renewable energy use and water-related impacts increase substantially. Depending on the production system, these impacts reach approximately 1.4–1.8 and 1.8–2.1 times the baseline values, respectively. Human-health impacts also approach or slightly exceed the baseline level.

The environmental burdens of Configuration 7 are primarily driven by the electricity required to produce hydrogen through water electrolysis. Under the assumed Swiss electricity supply, the high electricity demand shifts part of the environmental burden away from direct fossil CO_2 emissions toward upstream electricity-generation impacts. This result demonstrates that the favorable facility-level CO_2 balance of methanation does not necessarily translate into improvements across all life cycle impact categories.

4.4.2. Comparison across production systems

Despite differences in absolute impact magnitudes, similar configuration-level trends are observed for beer, whiskey, and combined production. In all three systems, heat-pump integration reduces the carbon footprint and non-renewable energy use relative to the baseline. Conversely, methanation consistently increases electricity-related resource and water impacts, illustrating a similar burden-shifting pattern across the three production systems.

Whiskey production has the highest baseline carbon footprint and non-renewable energy use, consistent with its greater thermal-energy demand. Beer production exhibits the highest baseline water-related impact, whereas human-health and ecosystem-quality results differ only marginally among the three systems. The combined-production system generally lies between the individual beer and whiskey cases for energy-related impact categories.

The broadly consistent environmental ranking of the configurations indicates that the principal trade-offs are governed by technology selection. However, process-specific characteristics remain important for individual impact categories, particularly those influenced by energy demand and water use. Overall, heat-pump-based configurations provide the most consistent multi-impact improvements, while methanation provides a carbon-footprint benefit at the expense of higher non-renewable energy use and water-related impacts.

Overall, the results identify heat-pump integration as the most robust decarbonization strategy across the three production systems. Heat recovery first reduces the residual utility demand, after which heat pumps efficiently upgrade available low-temperature heat and consistently reduce both operating energy requirements and life-cycle carbon impacts. Solar thermal integration becomes attractive under specific combinations of utility prices and carbon constraints but remains limited by temporal availability and temperature matching. Dregs valorization further reduces external energy consumption, while its storage is economically justified only when natural-gas substitution or electricity-price variability provides sufficient value. Methanation valorize biogenic CO_2 and can achieve a favorable facility-level carbon balance under low-electricity-price conditions; however, its high electricity requirement increases non-renewable energy use and water-related life-cycle impacts. These findings demonstrate that technology selection should account simultaneously for thermodynamic integration, economic conditions, temporal variability, and multi-impact environmental performance.

5. Conclusion

The brewery and distillery systems were modeled using DWSIM to evaluate beer, whiskey, and integrated production pathways. Heat integration analysis showed that appropriate selection of ΔT_{min} enables significant internal heat recovery, reducing heating and cooling demands by 50–75%.

Further integration using GCC and UGCC revealed that solar thermal systems provide moderate but highly seasonal reductions in natural gas consumption, whereas heat pump integration enables substantial improvements in energy efficiency and process electrification. Heat pumps reduced process CO_2 emissions by more than an order of magnitude compared to the baseline, demonstrating their central role in decarbonized utility systems.

Resource recovery from dregs via biogas production further reduced fossil CO_2 emissions and enabled partial self-sufficiency in heat and power supply. In contrast, CO_2 methanation offered limited feasibility due to its high electricity demand, despite its potential for synthetic fuel production and biogenic CO_2 utilization.

A techno-economic assessment across energy price and carbon tax scenarios identified heat pump-based systems as the most robust and consistently optimal solution. Hybrid solar–heat pump configurations become attractive under intermediate price conditions, while CO_2 utilization pathways are only viable in specific low-electricity, high-gas price scenarios. Overall, three operating regimes emerge: (i) heat pump-dominated low-cost systems, (ii) hybrid renewable integration under intermediate conditions, and (iii) carbon-constrained systems enabling CO_2 valorization.

Life cycle assessment confirms that heat pump-based configurations achieve the lowest environmental impacts across most categories. However, CO_2 methanation introduces trade-offs, increasing resource and water-related impacts due to high electricity consumption.

Overall, the study demonstrates that deep decarbonization of brewery and distillery systems is primarily driven by electrification through heat pumps, complemented by biomass valorization and, under specific conditions, solar thermal and CO_2 utilization pathways.

Raw material	Unit	Case A	Case B	Case C
Water	ton/h	13.4	8.7	12.6
Malt	ton/h	10.26	10.26	10.26
Soda	kg/h	133	133	133
Wastewater	m^3/h	39	84	61.5
Dregs	ton/h	2.7	2.7	2.7
Beer	m^3/h	98	-	49
Whiskey	m^3/h	-	5.8	2.9

Table A.10: Raw material flow rate for different cases

Table B.11: Capital cost assumptions for system components

Component	Lifetime	Cost	Reference
Solar thermal	20	500 (€/m^2)	[24]
Heat pump	20	750 (€/kW)	[25]
Alkaline Electrolyzer	30	1200 (€/kW)	[26]
Methanator	30	300 (€/kW)	[18]
Storing CH_4	30	4000 (€/(kg/h))	[27]
CO_2 storage	30	200 (€/(kg/h)	[28]
Thermal storage	30	74.4(€/m^3)	[29]
WWT	30	6000 (€/(m^3/day))	[30]
Biogas Engine	40	1850 (€/(kW))	[31]

6. Acknowledgments

Appendix A. Raw material consumption

Appendix B. Cost of technology

Appendix C. Share of technology in each region

Appendix C.1. Case 2: Production of beer

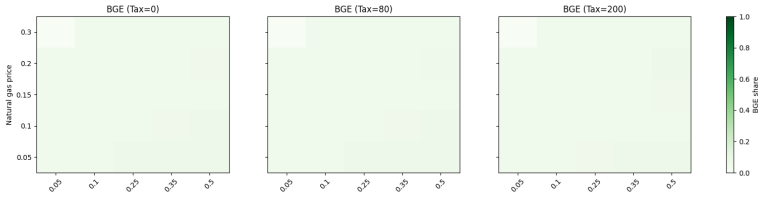


Figure C.20: Share of Biogas Engine in each scenario for beer production process

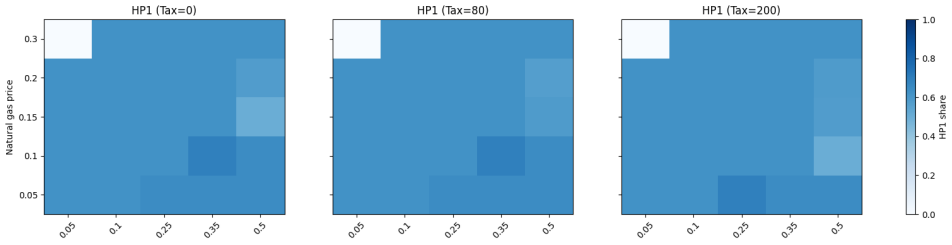


Figure C.21: Share of Heat-pump in each scenario for beer process

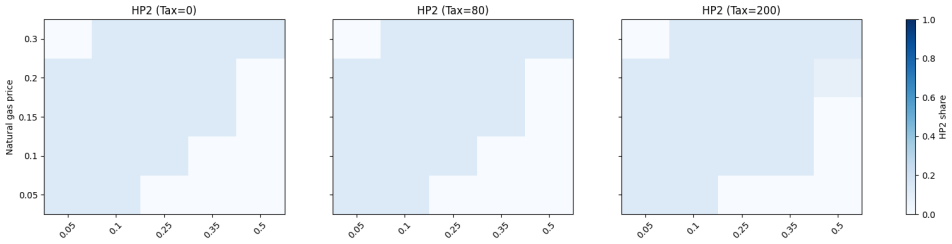


Figure C.22: Share of Heat-pump 2 in each scenario for beer production process

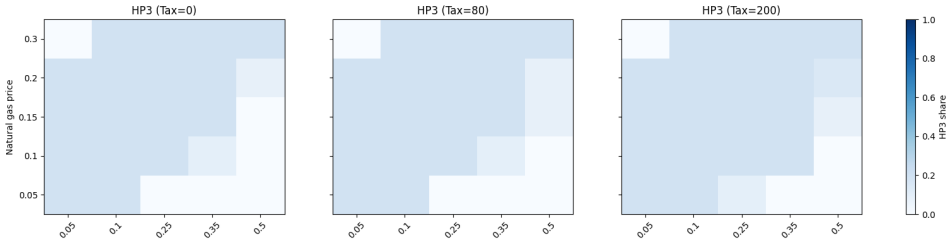


Figure C.23: Share of Heat-pump 3 in each scenario for beer production process

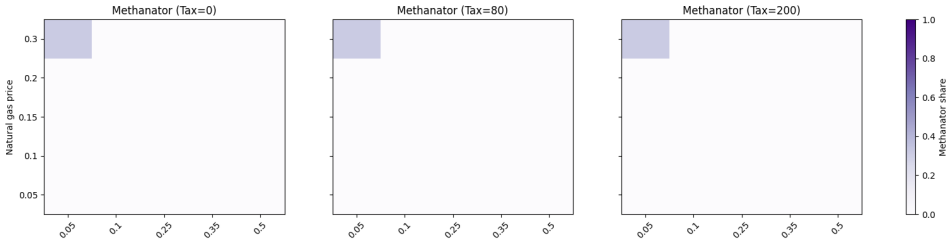


Figure C.24: Share of methanator process for beer production process

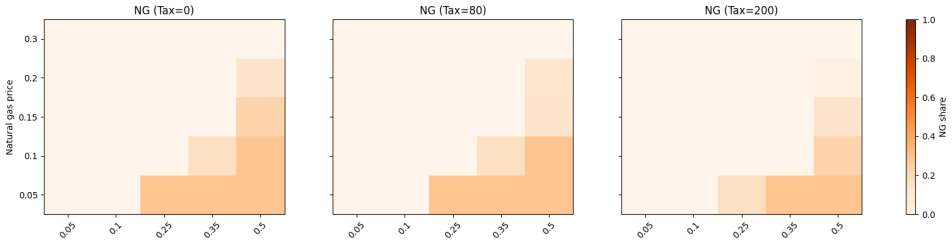


Figure C.25: Share of Natural Gas process for beer production process

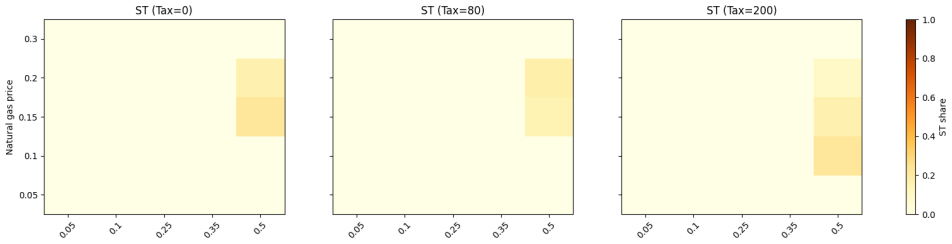


Figure C.26: Share of Solar thermal process for beer production process

Appendix C.2. Case 2: Production of whiskey

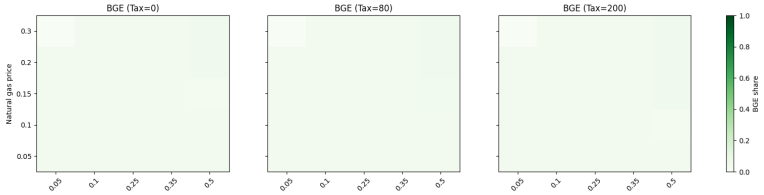


Figure C.27: Share of Biogas Engine in each scenario for whiskey production process

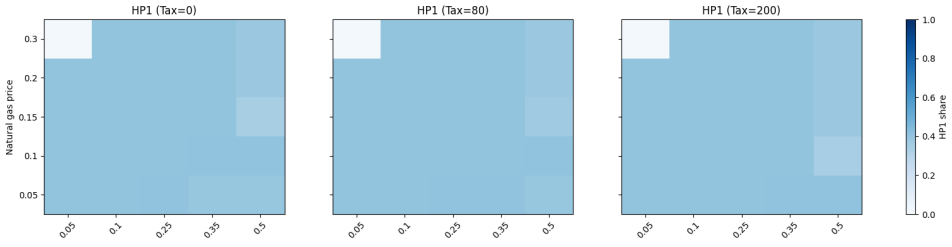


Figure C.28: Share of Heat-pump in each scenario for whiskey production process

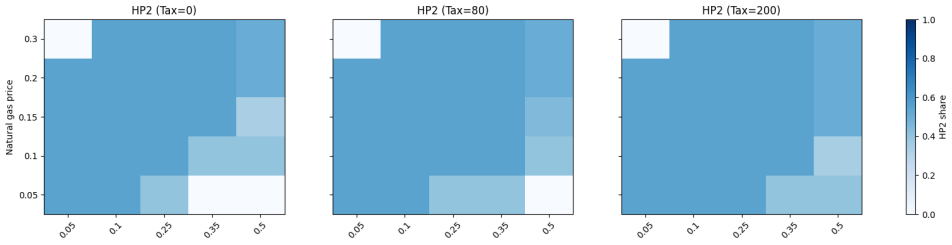


Figure C.29: Share of Heat-pump 2 in each scenario for whiskey production process

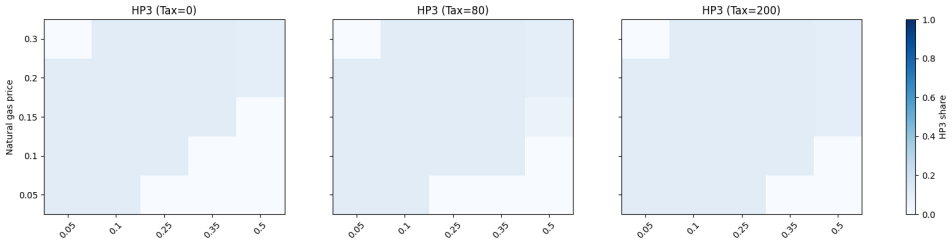


Figure C.30: Share of Heat-pump 3 in each scenario for whiskey production process

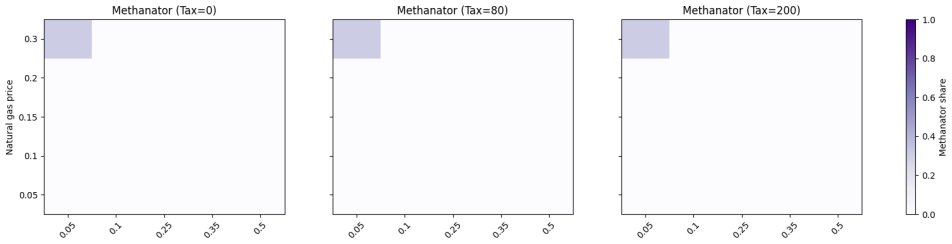


Figure C.31: Share of methanator process for whiskey production process

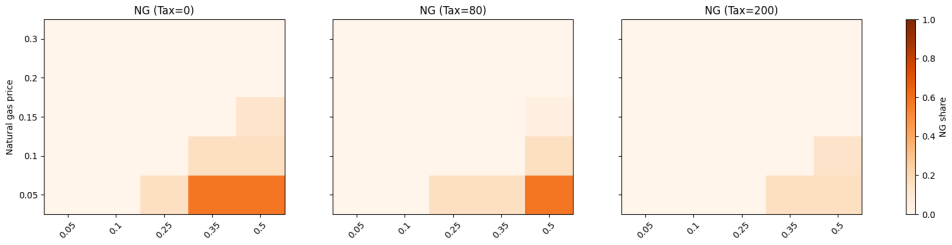


Figure C.32: hare of methanator process for Natural Gas production process

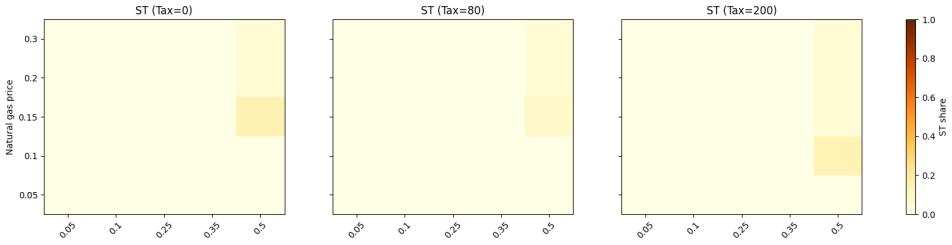


Figure C.33: hare of methanator process for Solar thermal production process

Appendix C.3. Case 3: Production of beer and whiskey

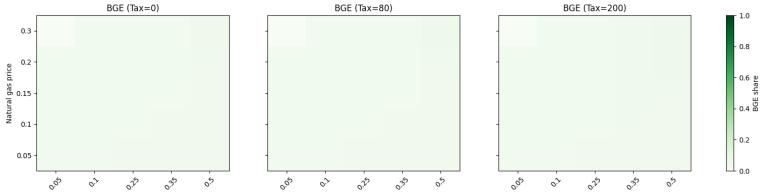


Figure C.34: Share of Biogas Engine in each scenario for case 3 production process

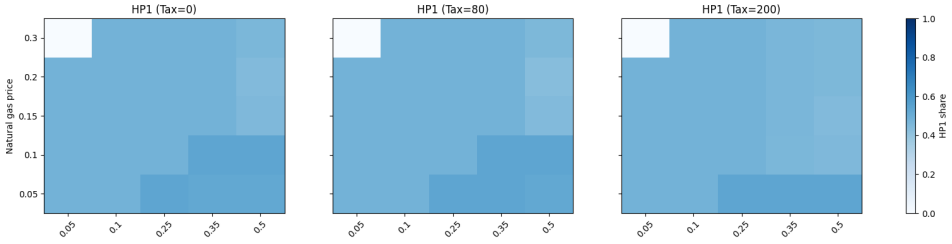


Figure C.35: Share of Heat-pump in each scenario for case 3 production process

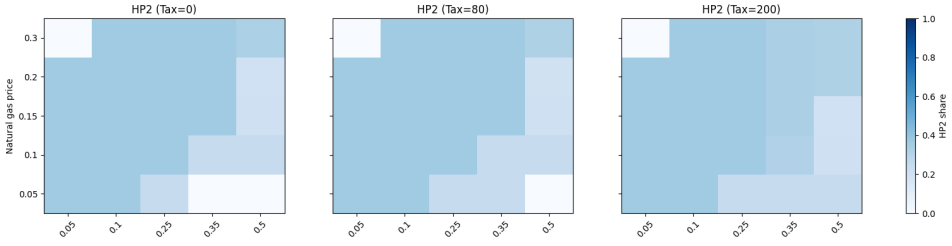


Figure C.36: Share of Heat-pump 2 in each scenario for case 3 production process

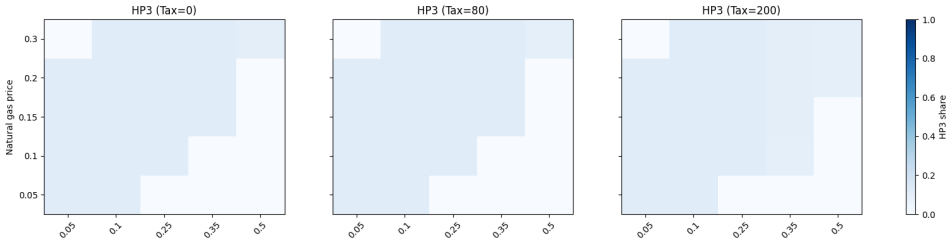


Figure C.37: Share of Heat-pump 3 in each scenario for case 3 production process

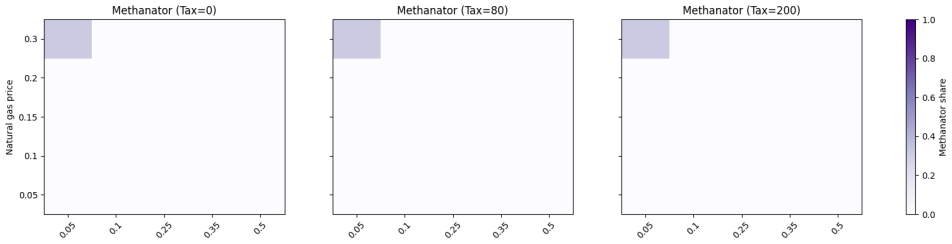


Figure C.38: Share of methanator process for case 3 production process

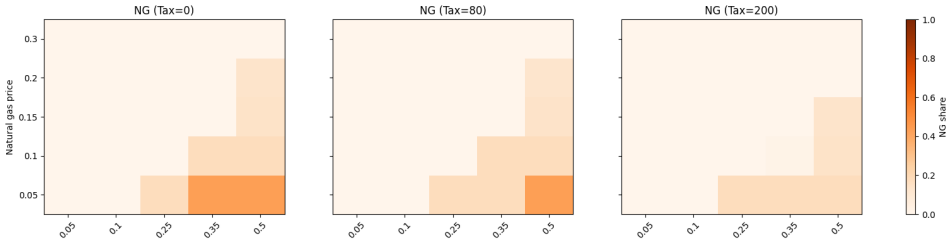


Figure C.39: Share of Natural gas process for case 3 production process

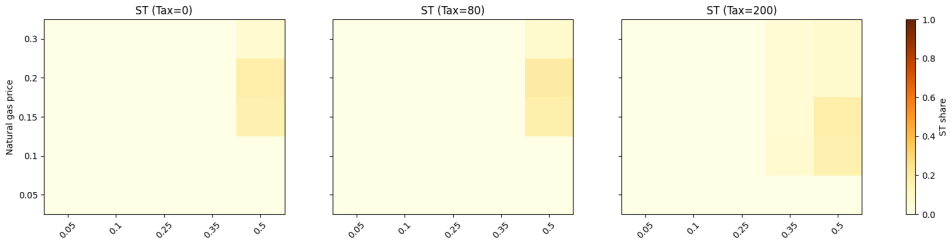


Figure C.40: Share of Biogas engine process for case 3 production process

Appendix D. Impact analysis for different configuration

Appendix D.0.1. Flow considered in across all the production process

Flows	Unit	Activity name	Reference product	Geography
Water	kg	market for tap water	tap water	CH
Malt	kg	market for barley grain, Swiss integrated production	barley grain, Swiss integrated production	CH
Soda	kg	market for soda ash, light	soda ash, light	RER
Wastewater	m^3	market for wastewater from anaerobic digestion of whey	wastewater from anaerobic digestion of whey	CH
NG	MJ	market for heat, district or industrial, natural gas	heat, district or industrial, natural gas	CH
Electricity	kWh	market for electricity, medium voltage	electricity, medium voltage	CH
Dregs	kg	market for green liquor dregs	green liquor dregs	RER
Heatpump	unit	heat pump production, heat and power co-generation unit, 160kW electrical	heat pump, heat and power co-generation unit, 160kW electrical	RER
Anaerobic digester	unit	anaerobic digestion plant construction, for biowaste	anaerobic digestion plant, for biowaste	CH
Engine	unit	heat and power co-generation unit construction, 160kW electrical, common components for heat+electricity	heat and power co-generation unit, 160kW electrical, common components for heat+electricity	RER
Solar Thermal	unit	concentrated solar power plant construction, solar thermal parabolic trough, 50 MW	concentrated solar power plant, solar thermal parabolic trough, 50 MW	RoW
Electrolyzer	unit	market for fuel cell, stack solid oxide, 125kW electrical, future	fuel cell, stack solid oxide, 125kW electrical, future	GLO
Methanator	unit	natural gas processing plant production	natural gas processing plant	GLO
Ash	kg	treatment of wood ash mixture, pure, sanitary landfill	wood ash mixture, pure	CH
Natural Gas Society	MJ	market for heat, district or industrial, natural gas	heat, district or industrial, natural gas	CH

Table D.12: Foreground and background flows considered in the conversion process

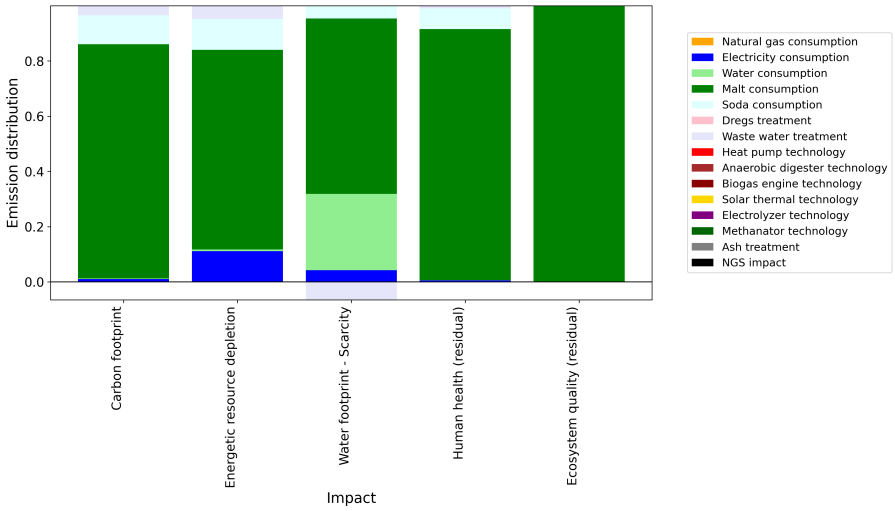


Figure D.41: Share of emission for beer in configuration 2

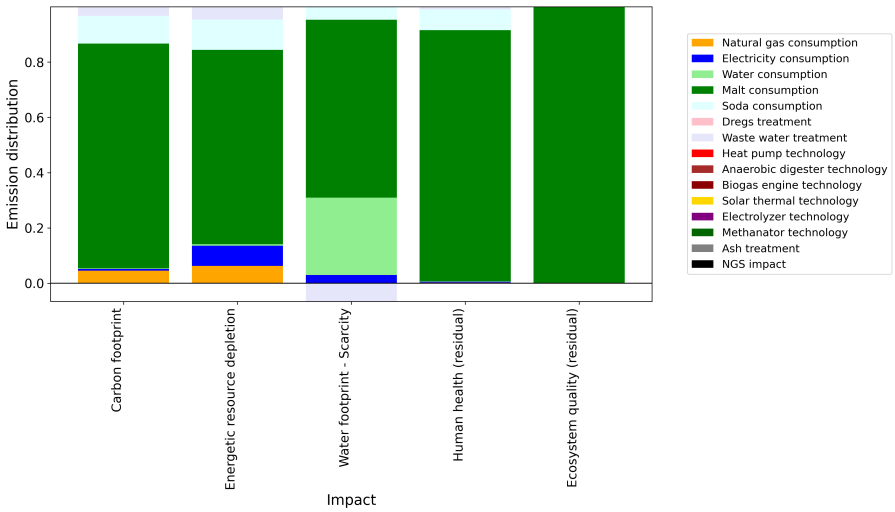


Figure D.42: Share of emission for beer in configuration 3

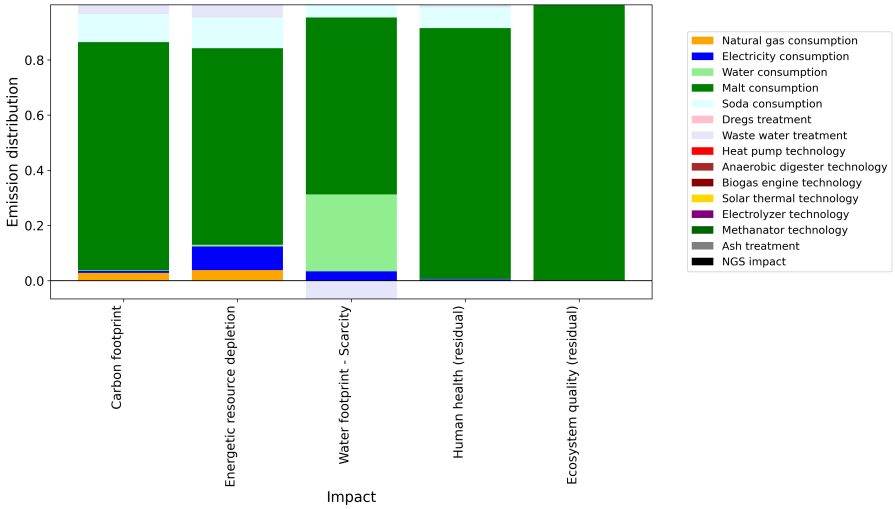


Figure D.43: Share of emission for beer in configuration 4

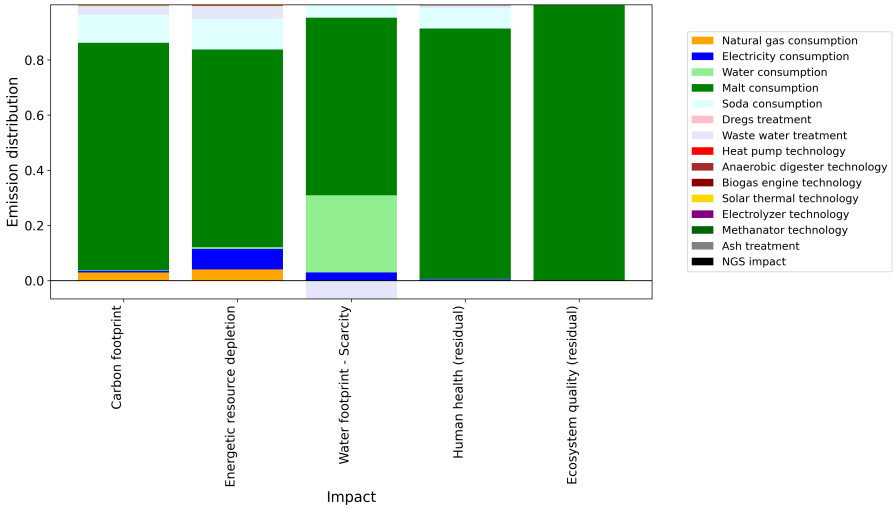


Figure D.44: Share of emission for beer in configuration 5

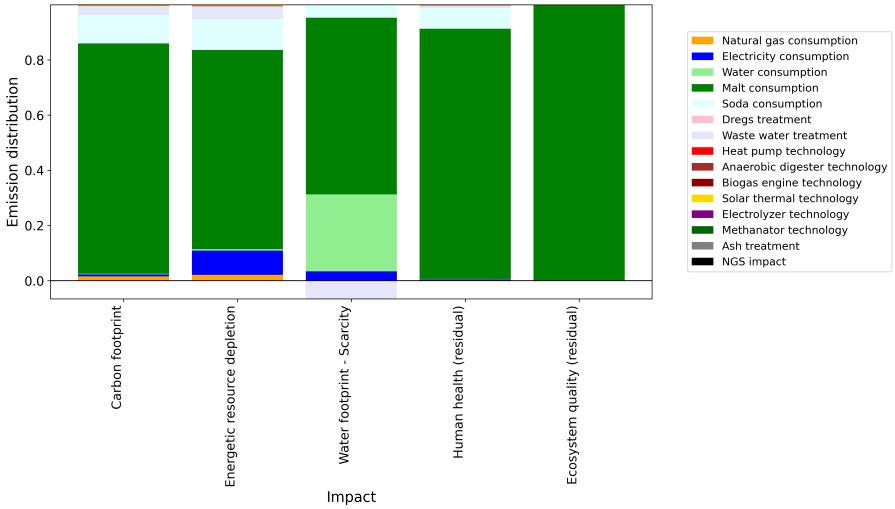


Figure D.45: Share of emission for beer in configuration 6

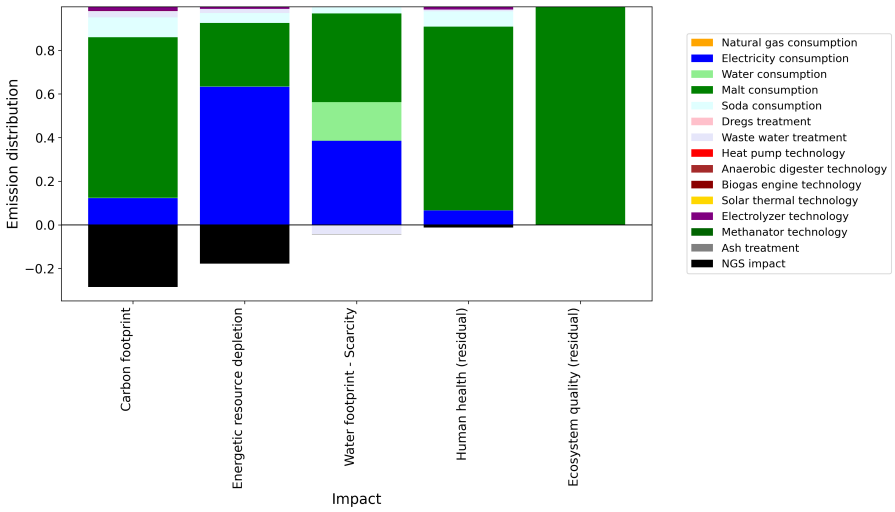


Figure D.46: Share of emission for beer in configuration 7

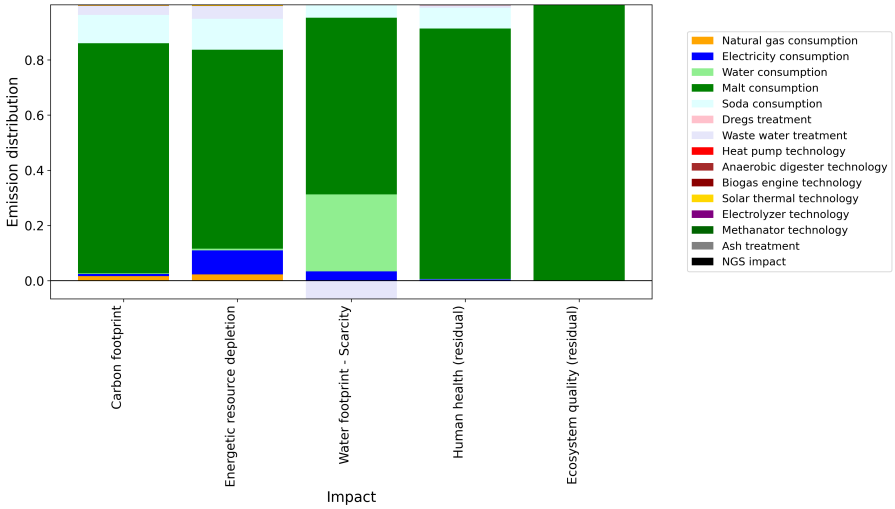


Figure D.47: Share of emission for beer in configuration 8

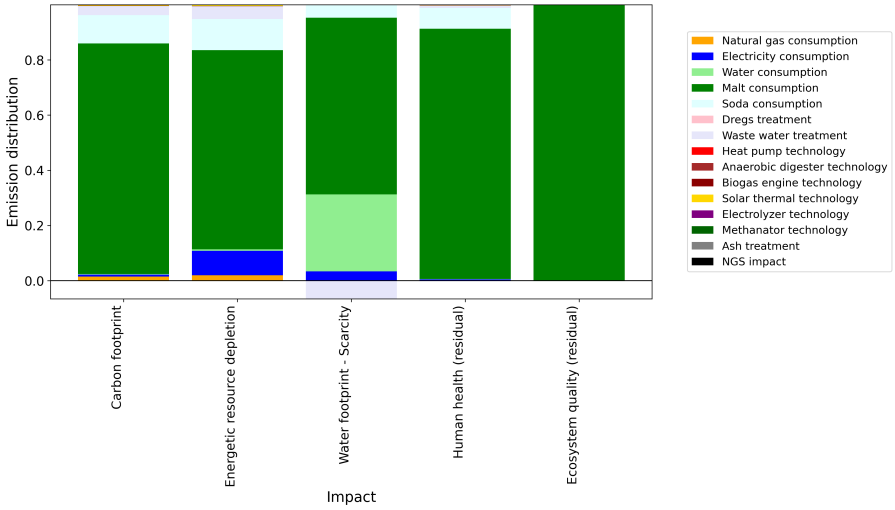


Figure D.48: Share of emission for beer in configuration 9

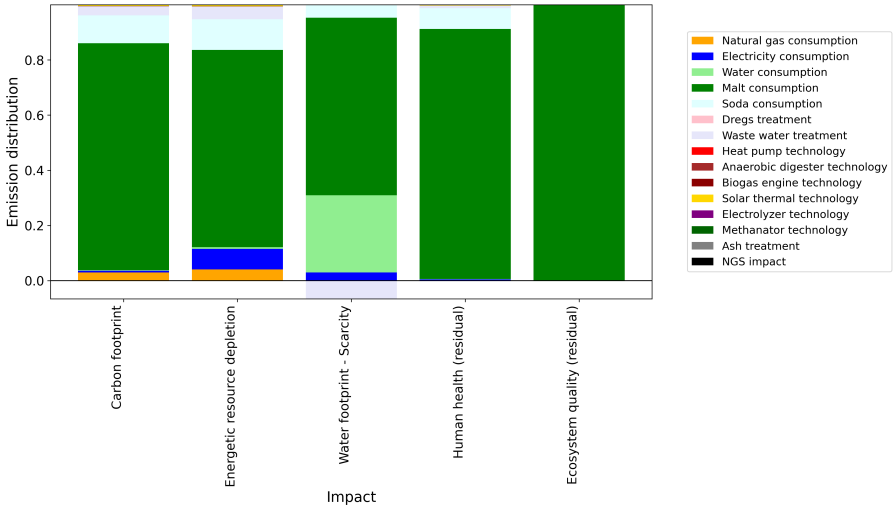


Figure D.49: Share of emission for beer in configuration 10

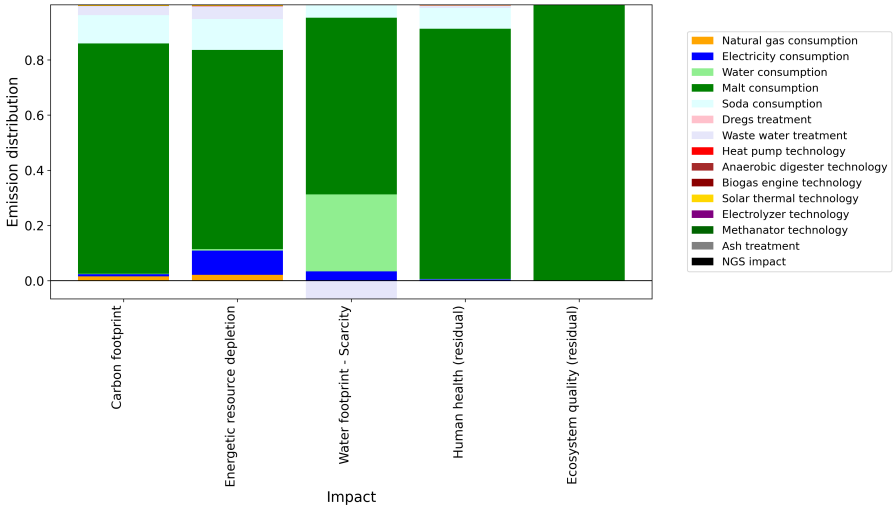


Figure D.50: Share of emission for beer in configuration 11

Appendix D.0.3. Case 2: Whiskey production process

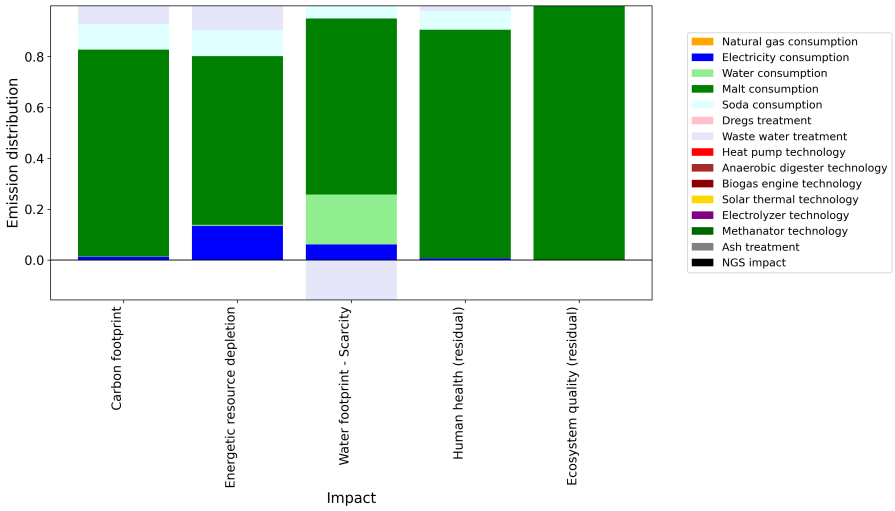


Figure D.51: Share of emission in configuration 2 for whiskey production process

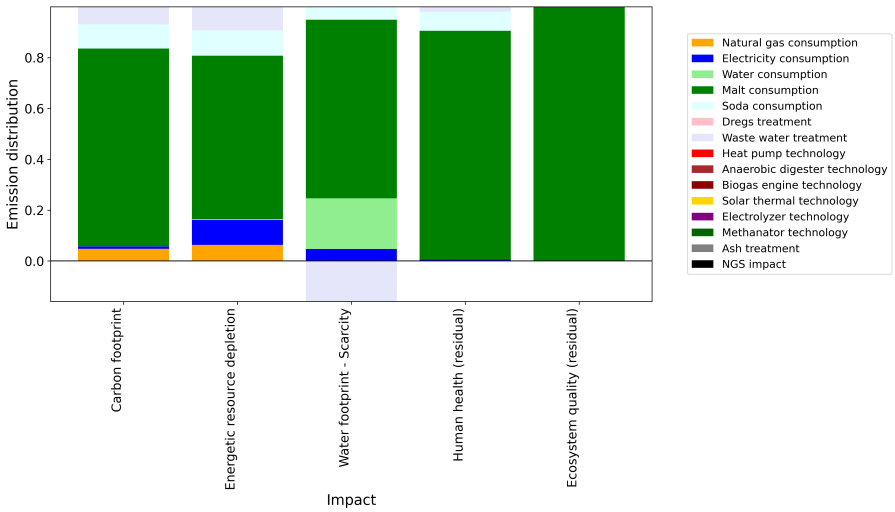


Figure D.52: Share of emission in configuration 3 for whiskey production process

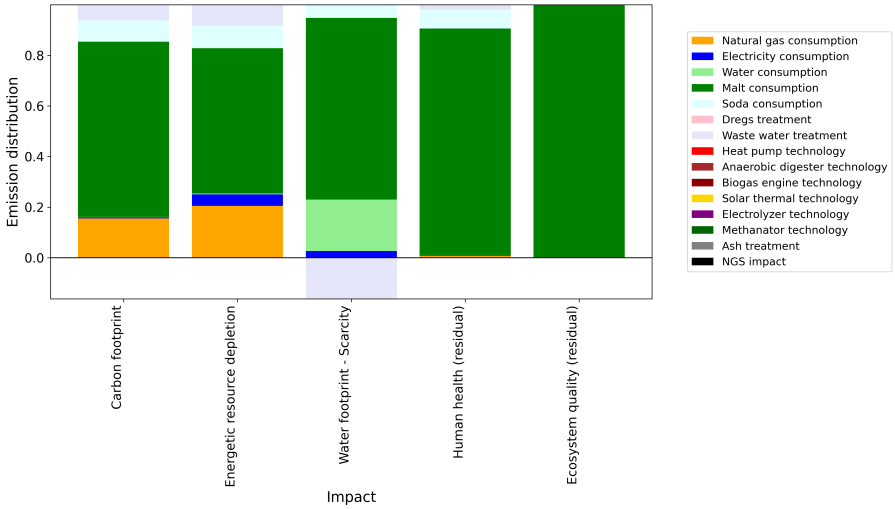


Figure D.53: Share of emission in configuration 4 for whiskey production process

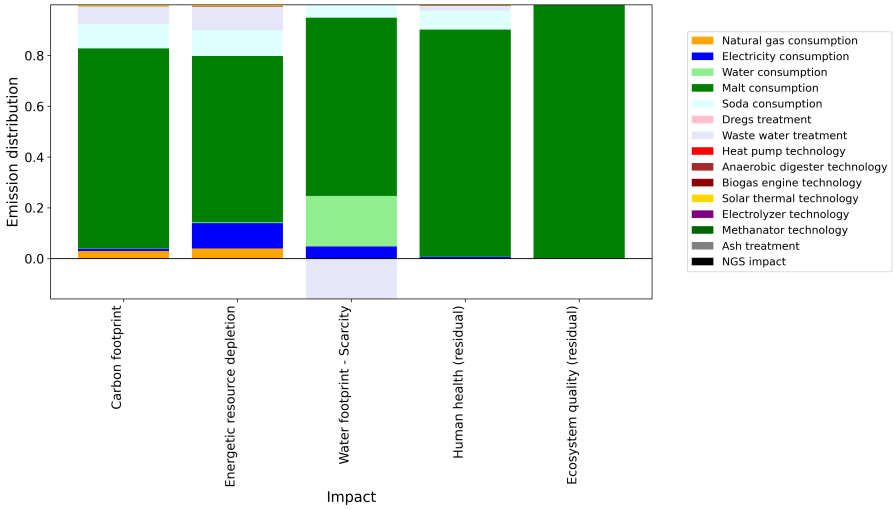


Figure D.54: Share of emission in configuration 5 for whiskey production process

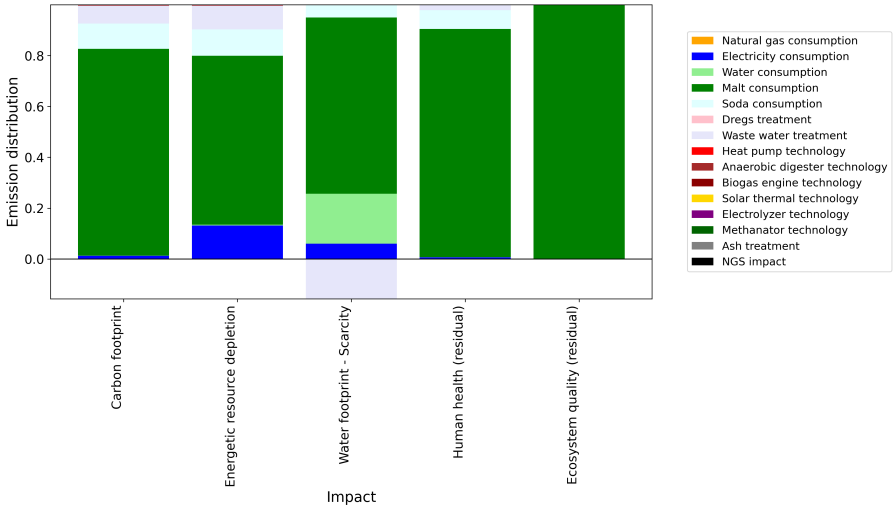


Figure D.55: Share of emission in configuration 6 for whiskey production process

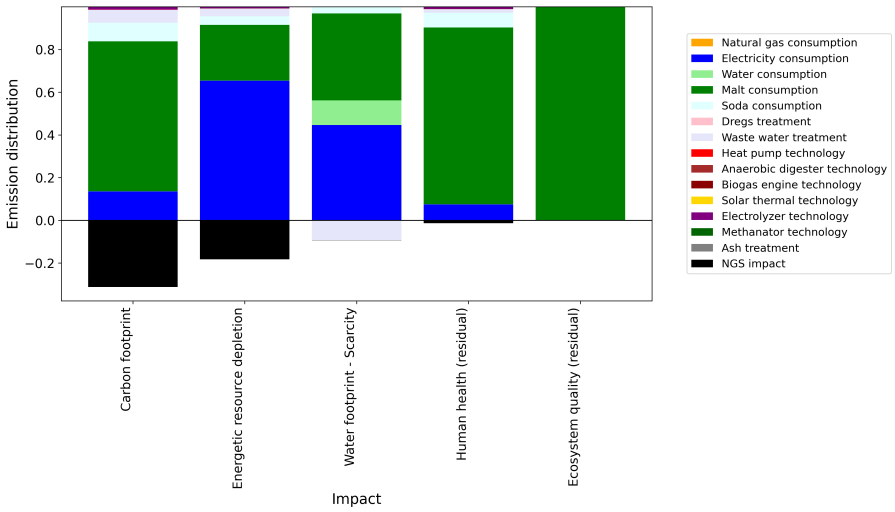


Figure D.56: Share of emission in configuration 7 for whiskey production process

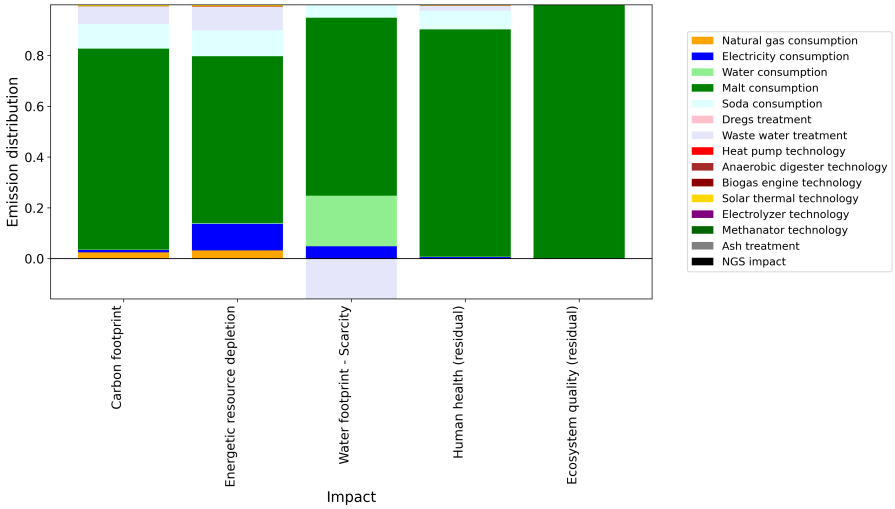


Figure D.57: Share of emission in configuration 8 for whiskey production process

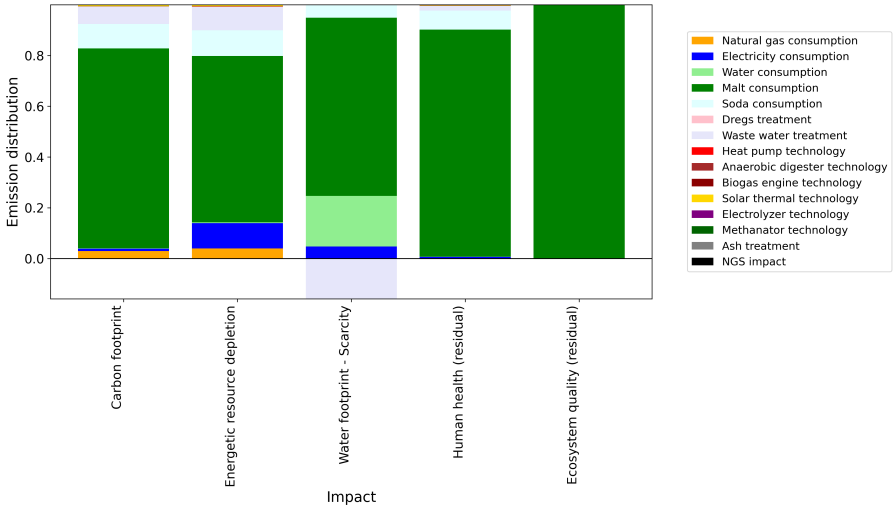


Figure D.58: Share of emission in configuration 9 for whiskey production process

Appendix D.0.4. Case 3: Combined production process

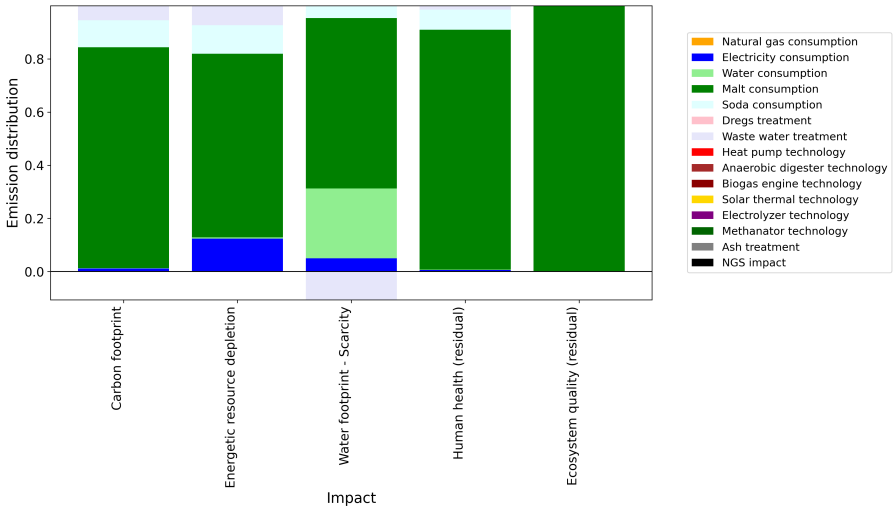


Figure D.59: Share of emission in configuration 2 for combined production process

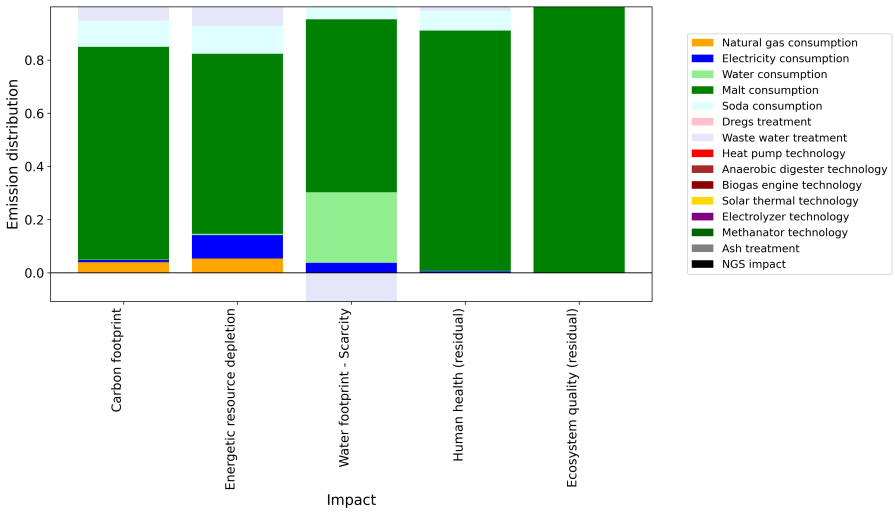


Figure D.60: Share of emission in configuration 3 for combined production process

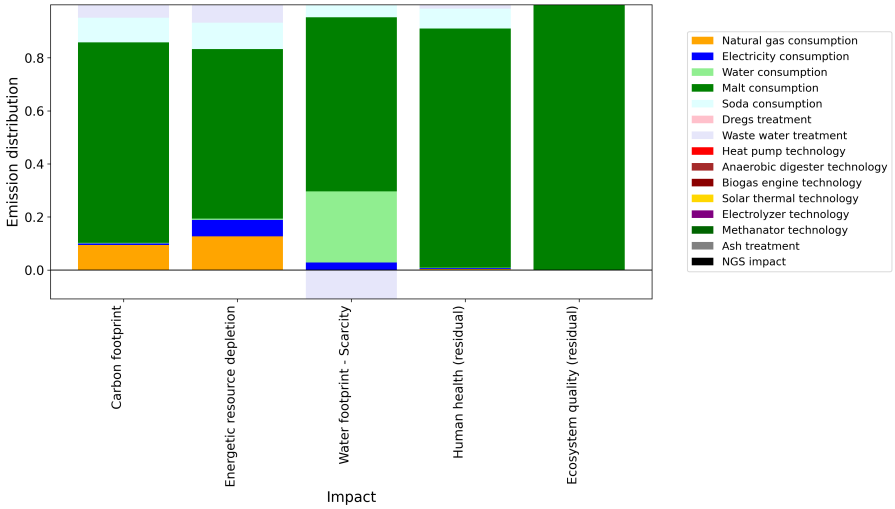


Figure D.61: Share of emission in configuration 4 for combined production process

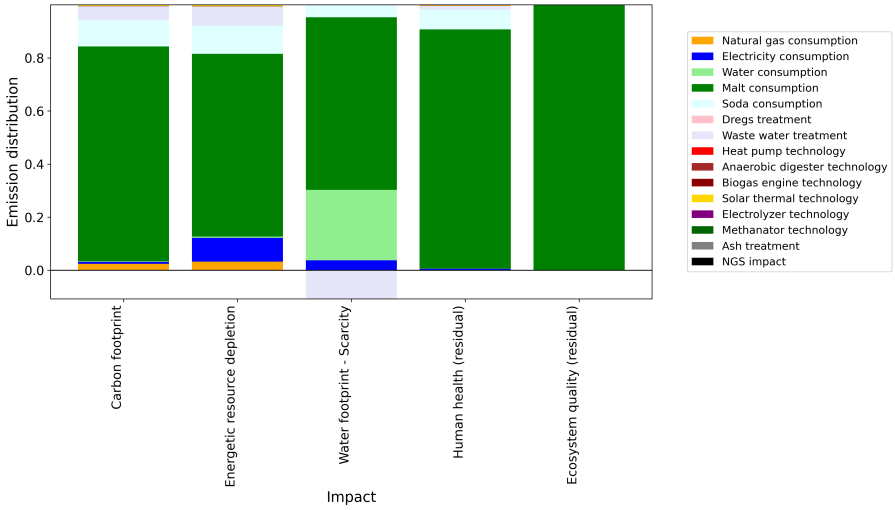


Figure D.62: Share of emission in configuration 5 for combined production process

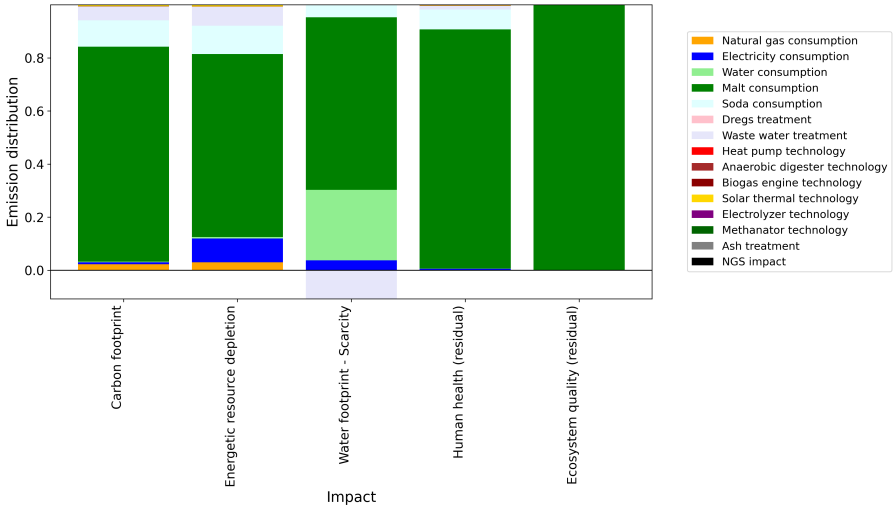


Figure D.63: SShare of emission in configuration 6 for combined production process

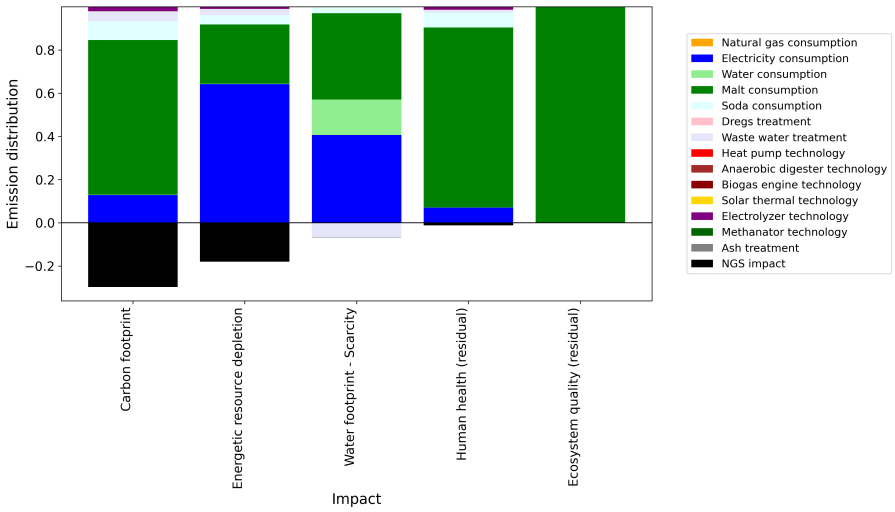


Figure D.64: Share of emission in configuration 7 for combined production process

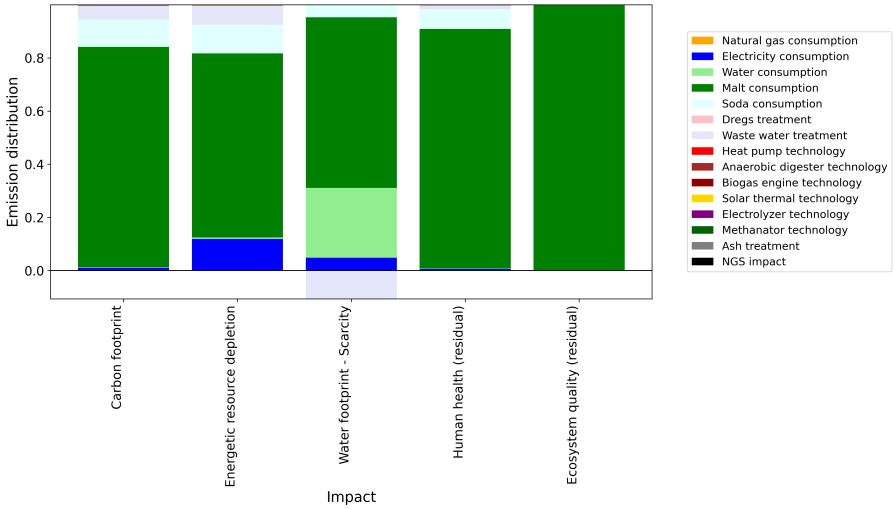


Figure D.65: Share of emission in configuration 8 for combined production process

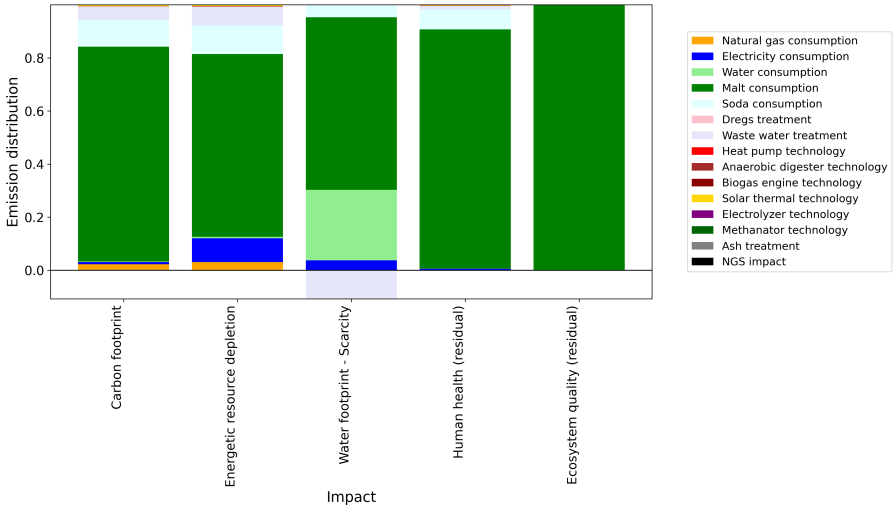


Figure D.66: Share of emission in configuration 9 for combined production process

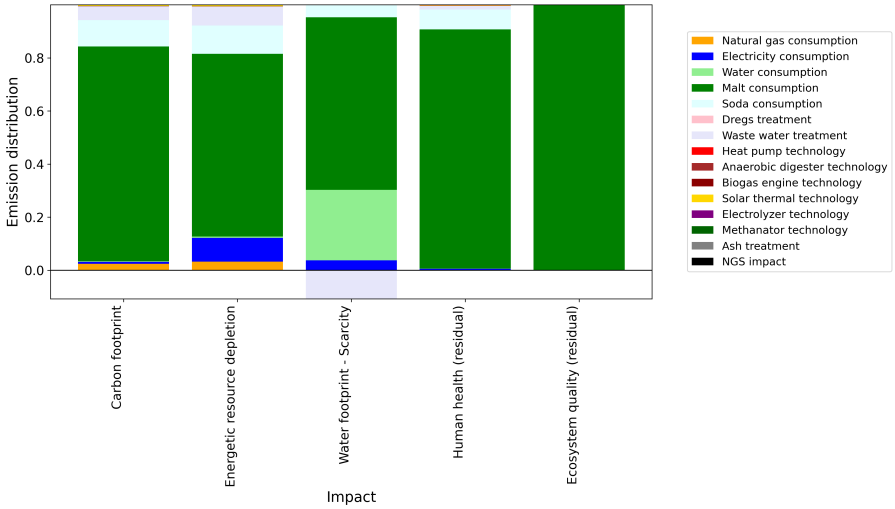


Figure D.67: Share of emission in configuration 10 for combined production process

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